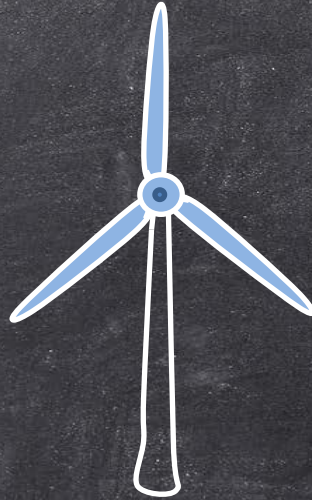


Rotor and Wake Aerodynamics

Actuator disk momentum theory

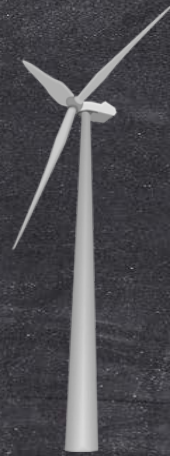
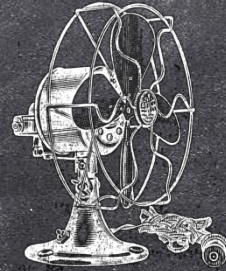


How do we build a model for rotors?



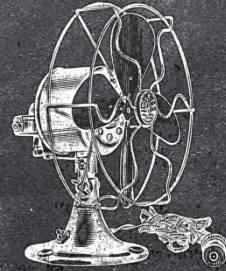
Why do we want to model rotors?

- To make wind flow
- To create propulsion
- To convert power



Which equations to use?

- To make wind flow
 - Continuity equation
- To create propulsion
 - Momentum equation
- To convert power
 - Energy equation



➤ Continuity equation

$$\frac{\partial}{\partial t} \iiint_V \rho dV + \iint_S \rho \vec{U} \cdot \vec{n} dS = 0$$

➤ Momentum equation

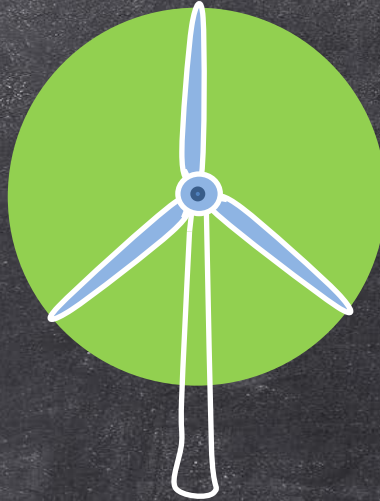
$$\frac{\partial}{\partial t} \iiint_V \rho \vec{U} dV + \iint_S \rho \vec{U} (\vec{U} \cdot \vec{n} dS) = - \iint_S p \vec{n} dS + \iiint_V \rho \vec{f} dV + F_{viscous}$$

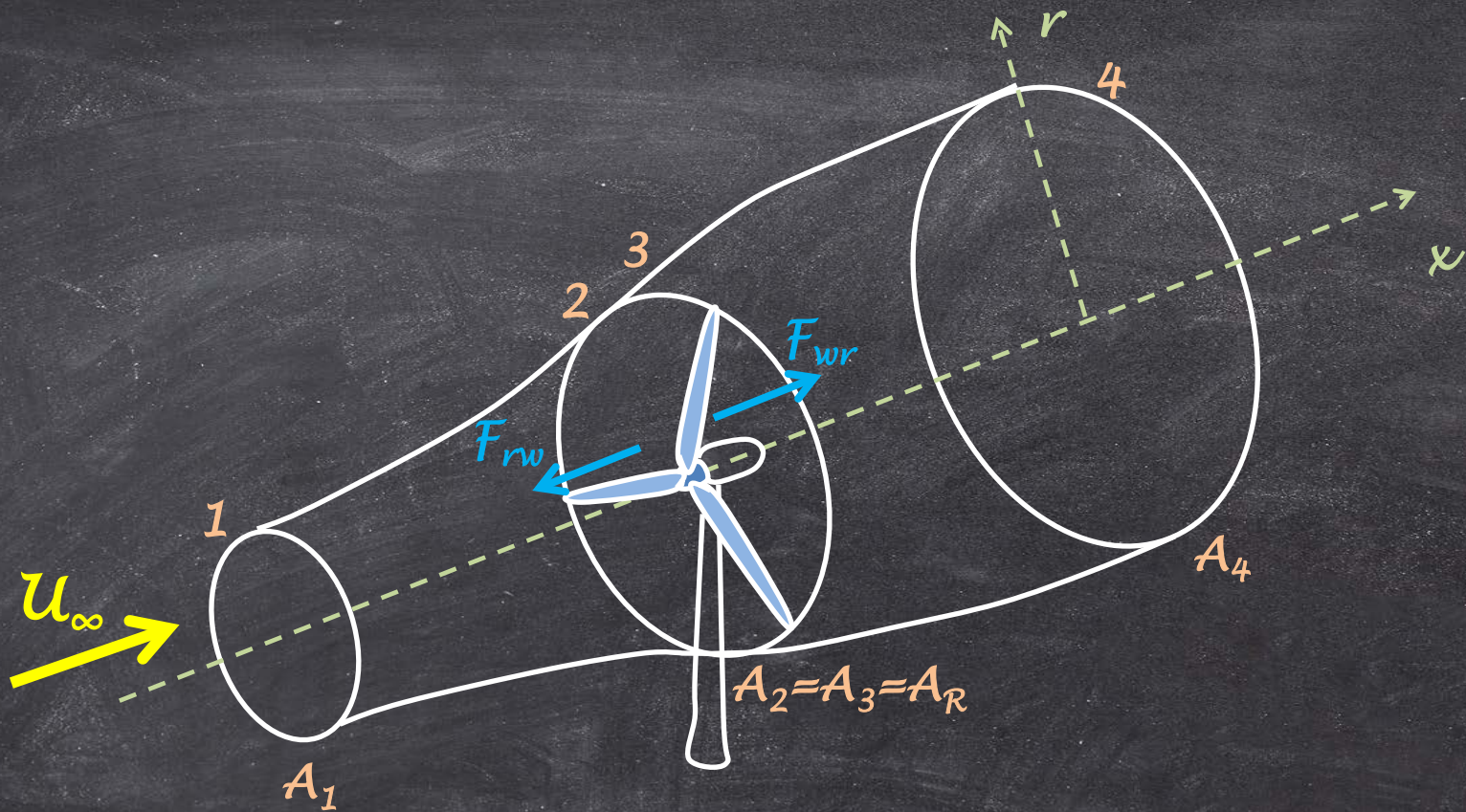
➤ Energy equation

$$\frac{\partial}{\partial t} \iiint_V \left(\rho e + \rho \frac{U^2}{2} \right) dV + \iint_S \left(\rho e + \rho \frac{U^2}{2} \right) (\vec{U} \cdot \vec{n} dS) =$$

$$\frac{\partial}{\partial t} \iiint_V \dot{q} dV + \dot{Q}_{viscous} - \iint_S p (\vec{U} \cdot \vec{n} dS) + \iiint_V \rho \vec{f} \cdot \vec{U} dV + \dot{W}_{viscous}$$

Where do we apply these equations?





Our objective for today

- Derive the actuator disk model
 - equations for continuity, momentum and energy
 - relations between induction, loads and power



➤ Continuity equation

$$\frac{\partial}{\partial t} \iiint_V \rho dV + \iint_S \rho \vec{U} \cdot \vec{n} dS = 0$$

➤ Momentum equation

$$\frac{\partial}{\partial t} \iiint_V \rho \vec{U} dV + \iint_S \rho \vec{U} (\vec{U} \cdot \vec{n} dS) = - \iint_S p \vec{n} dS + \iiint_V \rho \vec{f} dV + F_{viscous}$$

➤ Energy equation

$$\frac{\partial}{\partial t} \iiint_V \left(\rho e + \rho \frac{U^2}{2} \right) dV + \iint_S \left(\rho e + \rho \frac{U^2}{2} \right) (\vec{U} \cdot \vec{n} dS) =$$

$$\frac{\partial}{\partial t} \iiint_V \dot{q} dV + \dot{Q}_{viscous} - \iint_S p (\vec{U} \cdot \vec{n} dS) + \iiint_V \rho \vec{f} \cdot \vec{U} dV + \dot{W}_{viscous}$$

➤ Continuity equation

Steady

$$\frac{\partial}{\partial t} \iiint_V \rho dV + \iint_S \rho \vec{U} \cdot \vec{n} dS = 0$$

➤ Momentum equation

$$\frac{\partial}{\partial t} \iiint_V \rho \vec{U} dV + \iint_S \rho \vec{U} (\vec{U} \cdot \vec{n} dS) = - \iint_S p \vec{n} dS + \iiint_V \rho \vec{f} dV + F_{viscous}$$

➤ Energy equation

$$\frac{\partial}{\partial t} \iiint_V \left(\rho e + \rho \frac{U^2}{2} \right) dV + \iint_S \left(\rho e + \rho \frac{U^2}{2} \right) (\vec{U} \cdot \vec{n} dS) =$$

$$\frac{\partial}{\partial t} \iiint_V \rho dV + \dot{Q}_{viscous} - \iint_S p (\vec{U} \cdot \vec{n} dS) + \iiint_V \rho \vec{f} \cdot \vec{U} dV + \dot{W}_{viscous}$$

➤ Continuity equation

Steady, inviscid

$$\oiint_S \rho \vec{U} \cdot \vec{n} dS = 0$$

➤ Momentum equation

$$\oiint_S \rho \vec{U} (\vec{U} \cdot \vec{n} dS) = - \oiint_S p \vec{n} dS + \iiint_V \rho \vec{f} dV + F_{\text{ext}}$$

➤ Energy equation

$$\oiint_S \left(\rho e + \rho \frac{U^2}{2} \right) (\vec{U} \cdot \vec{n} dS) =$$

$$\dot{Q}_{\text{vis}} - \oiint_S p (\vec{U} \cdot \vec{n} dS) + \iiint_V \rho \vec{f} \cdot \vec{U} dV + \dot{W}_{\text{ext}}$$



➤ Continuity equation

Steady, inviscid, incompressible

$$\oint_S \rho \vec{U} \cdot \vec{n} dS = 0$$

➤ Momentum equation

$$\oint_S \rho \vec{U} (\vec{U} \cdot \vec{n} dS) = - \oint_S p \vec{n} dS + \iiint_V \rho \vec{f} dV$$

➤ Energy equation

$$\oint_S \left(\rho e + \rho \frac{U^2}{2} \right) (\vec{U} \cdot \vec{n} dS) = - \oint_S p (\vec{U} \cdot \vec{n} dS) + \iiint_V \rho \vec{f} \cdot \vec{U} dV$$



➤ Continuity equation

$$\oiint_S \vec{U} \cdot \vec{n} dS = 0$$

Steady, inviscid, incompressible,
constant internal energy

➤ Momentum equation

$$\oiint_S \rho \vec{U} (\vec{U} \cdot \vec{n} dS) = - \oiint_S p \vec{n} dS + \iiint_V \rho \vec{f} dV$$

➤ Energy equation

$$\oiint_S \left(\cancel{\rho e} + \rho \frac{U^2}{2} \right) (\vec{U} \cdot \vec{n} dS) =$$

$$\oiint_S p (\vec{U} \cdot \vec{n} dS) + \iiint_V \rho \vec{f} \cdot \vec{U} dV$$



➤ Continuity equation

$$\oiint_S \vec{U} \cdot \vec{n} dS = 0$$

Steady, inviscid, incompressible,
constant internal energy

➤ Momentum equation

$$\oiint_S \rho \vec{U} (\vec{U} \cdot \vec{n} dS) = - \oiint_S p \vec{n} dS + \iiint_V \rho \vec{f} dV$$

➤ Energy equation

$$\oiint_S \left(p + \rho \frac{U^2}{2} \right) (\vec{U} \cdot \vec{n} dS) = \iiint_V \rho \vec{f} \cdot \vec{U} dV$$

➤ Continuity equation

$$\oiint_S \vec{U} \cdot \vec{n} dS = 0$$

Steady, inviscid, incompressible,
constant internal energy,
Bernoulli's equation

➤ Momentum equation

$$\oiint_S \rho \vec{U} (\vec{U} \cdot \vec{n} dS) = - \oiint_S p \vec{n} dS + \iiint_V \rho \vec{f} dV$$

➤ Bernoulli's equation

$$p + \rho \frac{U^2}{2} = \text{constant}$$



➤ Continuity equation

$$\oiint_S \vec{U} \cdot \vec{n} dS = 0$$

We need only to evaluate pressure and velocity: p and U

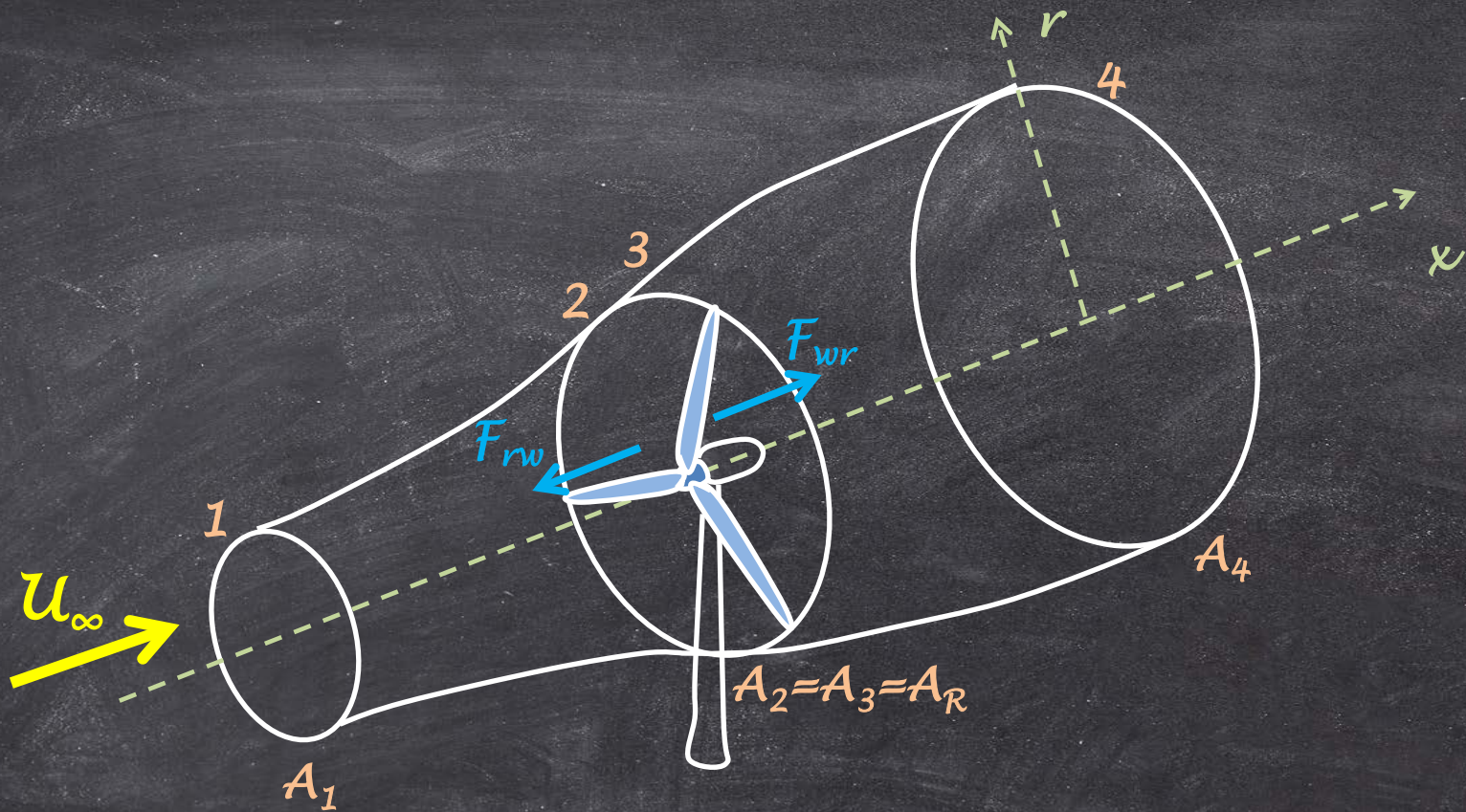
➤ Momentum equation

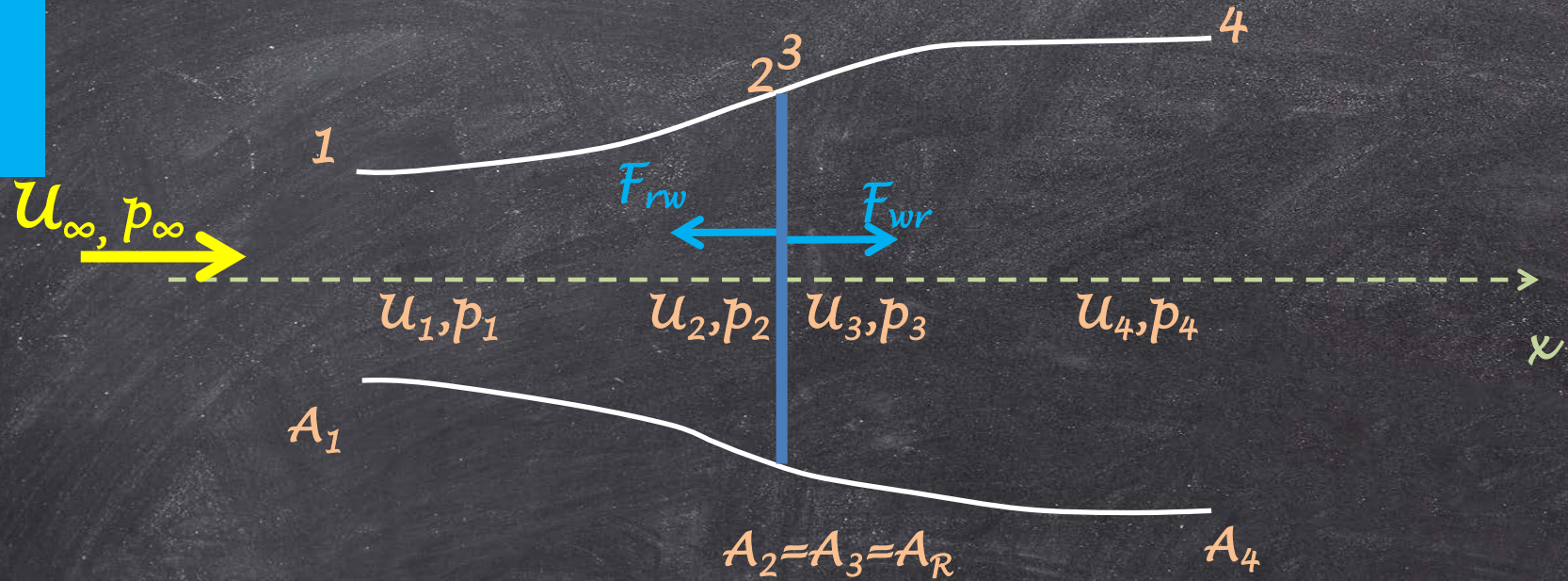
$$\oiint_S \rho \vec{U} (\vec{U} \cdot \vec{n} dS) = - \oiint_S p \vec{n} dS + \iiint_V \rho \vec{f} dV$$

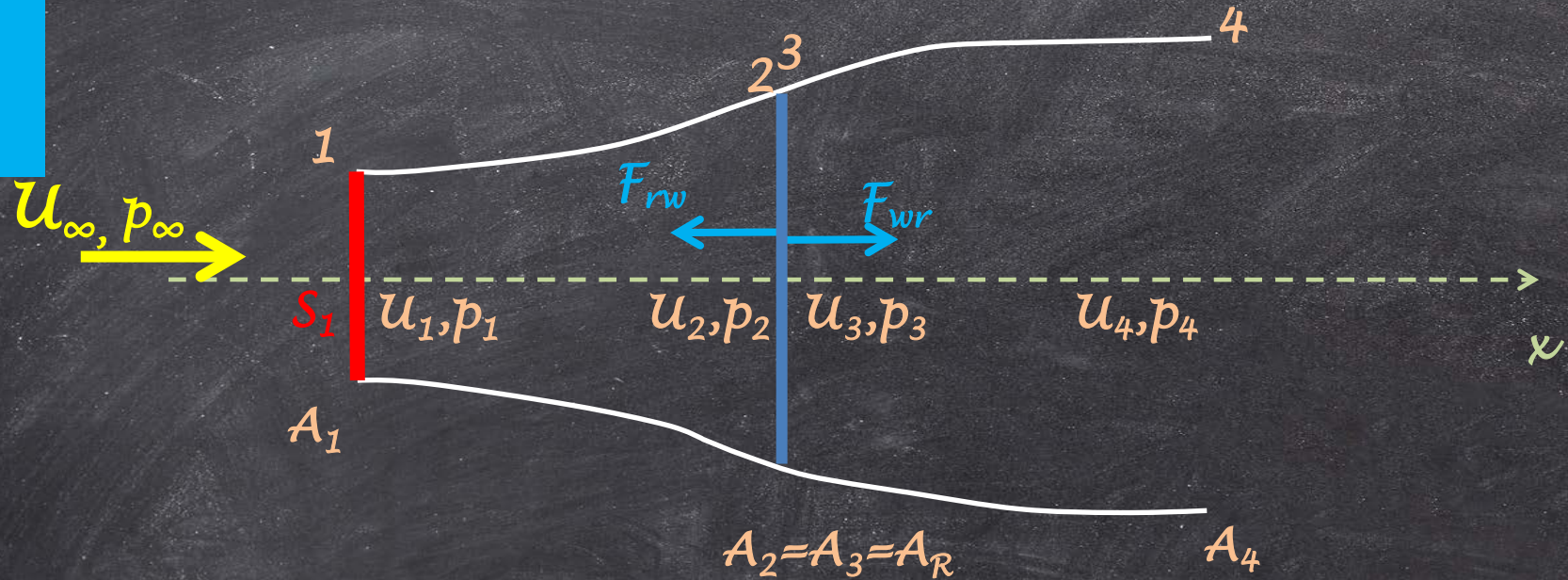
➤ Bernoulli's equation

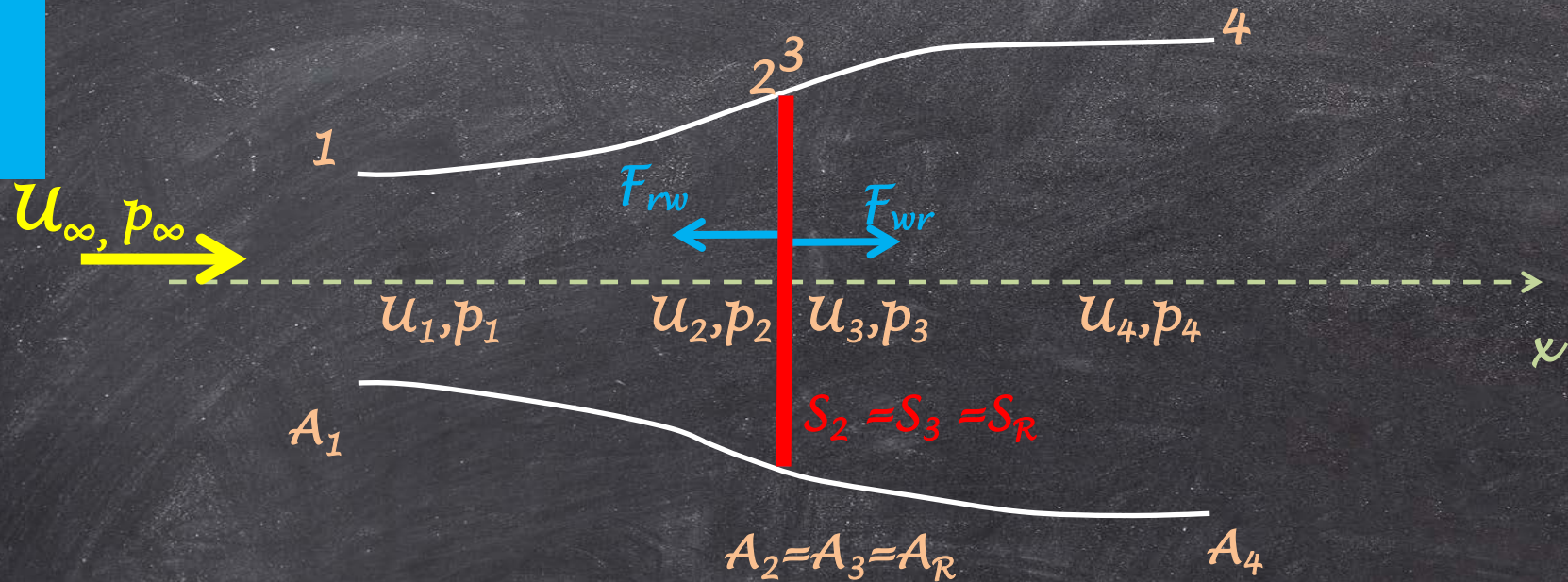
$$p + \rho \frac{U^2}{2} = \text{constant}$$

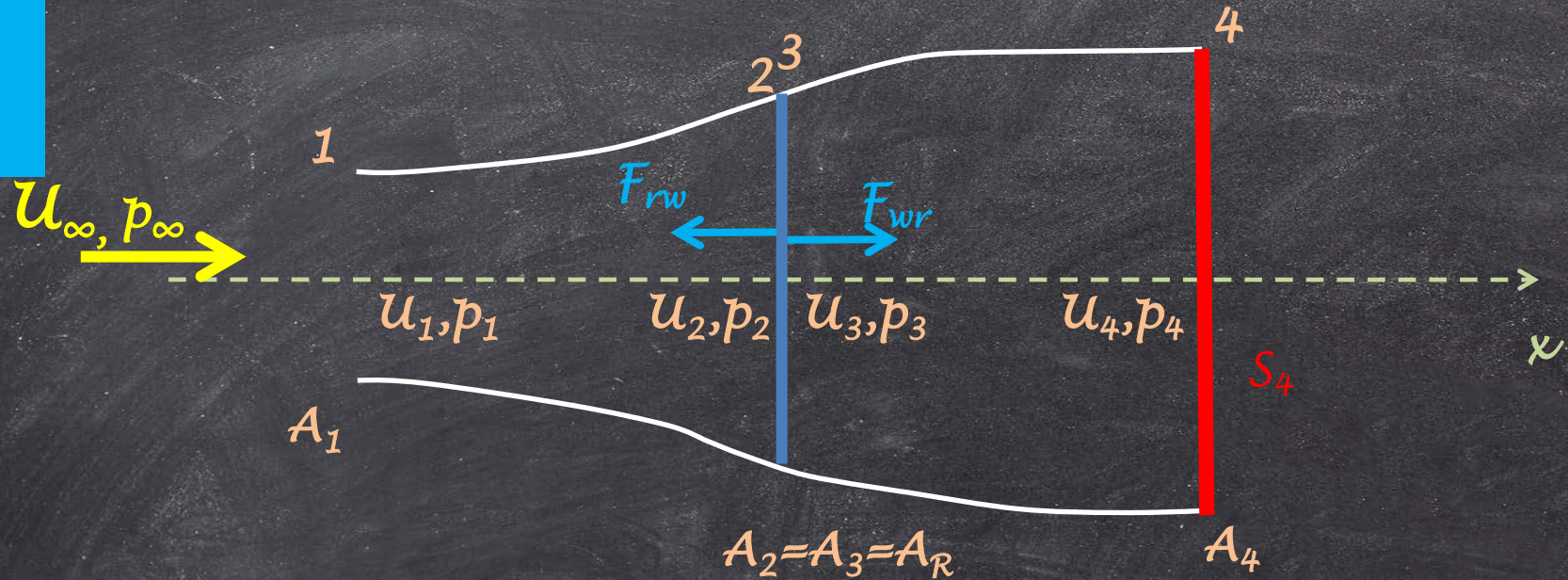


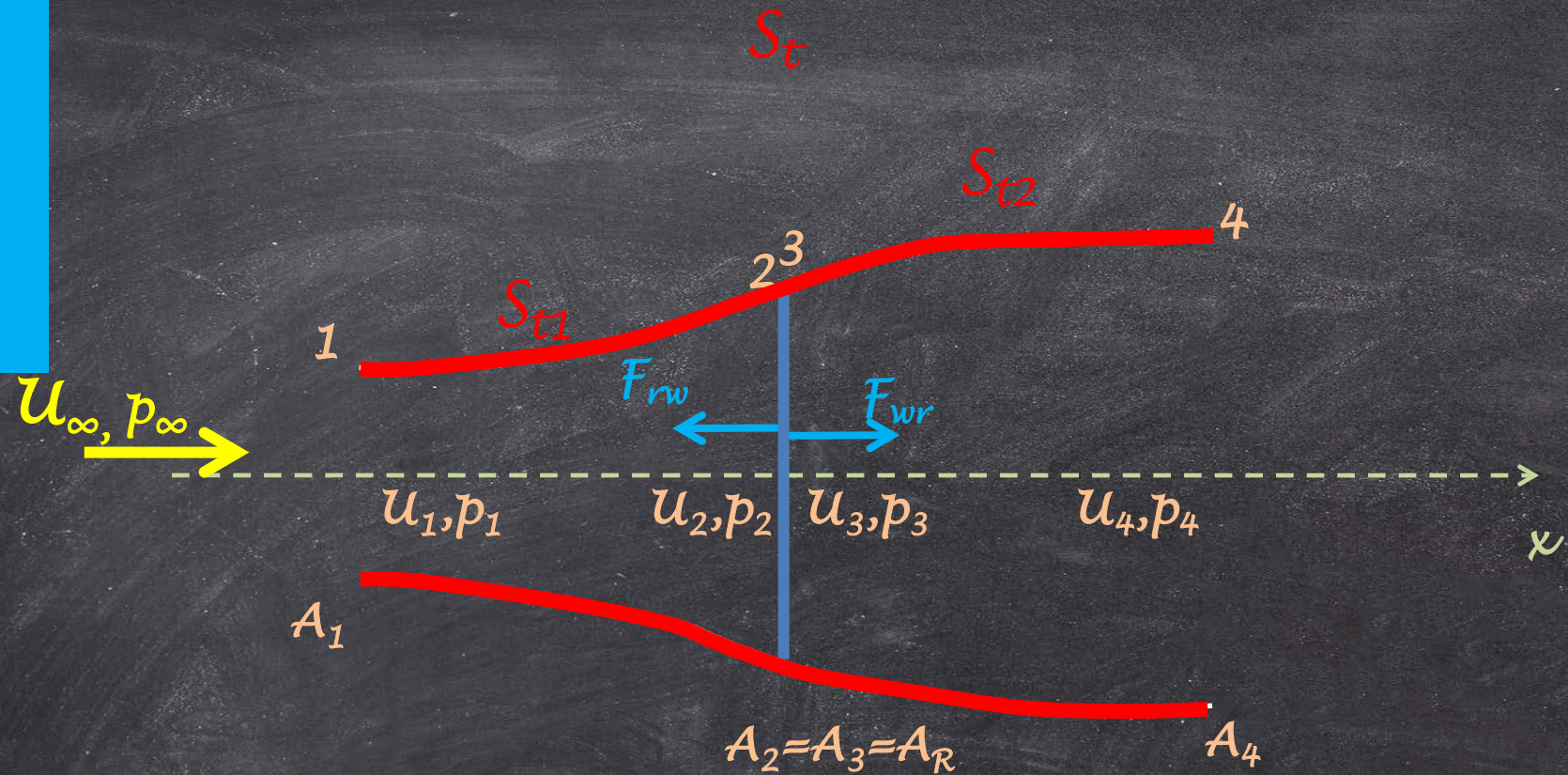


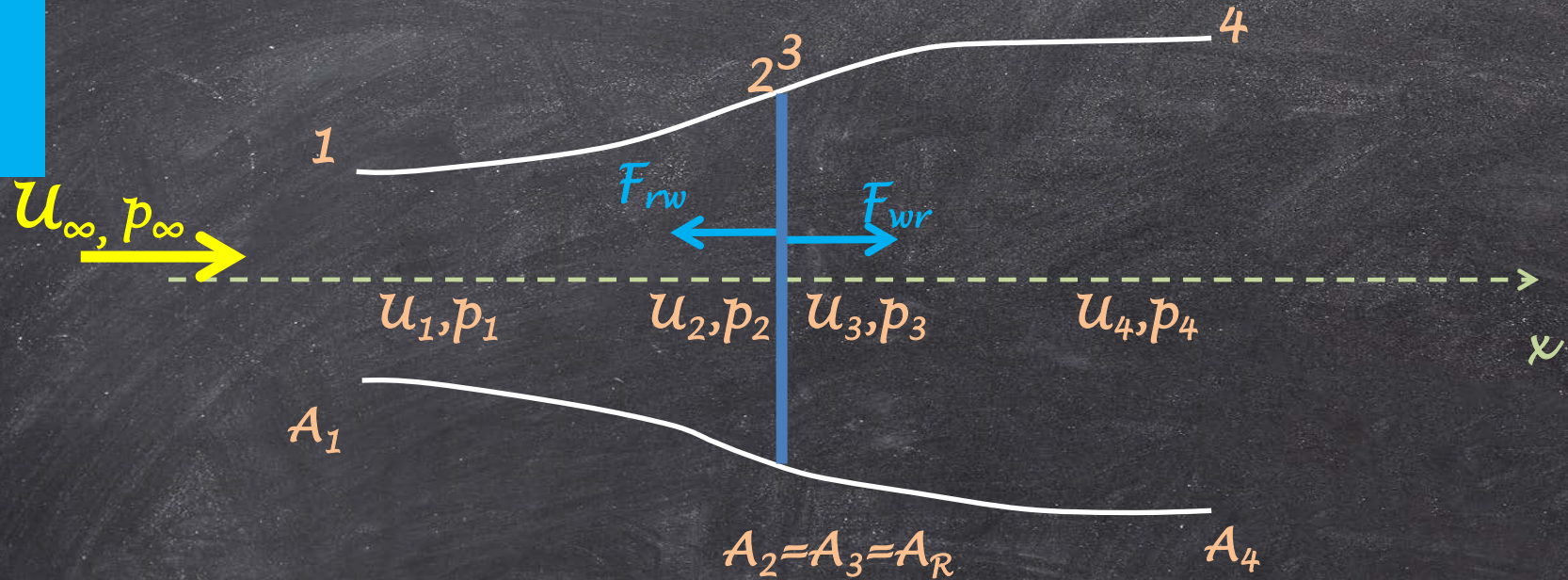












Setting the continuity equation

- We will define the continuity equation
- If you want to try to do it by yourself, pause the video now.

➤ Continuity equation

$$\oiint_S \vec{U} \cdot \vec{n} dS = 0$$

$$\oiint_{S_1} \vec{U} \cdot \vec{n} dS + \oiint_{S_t} \vec{U} \cdot \vec{n} dS + \oiint_{S_4} \vec{U} \cdot \vec{n} dS = 0$$

$$\oiint_{S_1} \vec{U} \cdot \vec{n} dS + \oiint_{S_{t1}} \vec{U} \cdot \vec{n} dS + \oiint_{S_2} \vec{U} \cdot \vec{n} dS = 0$$

$$\oiint_{S_3} \vec{U} \cdot \vec{n} dS + \oiint_{S_{t2}} \vec{U} \cdot \vec{n} dS + \oiint_{S_4} \vec{U} \cdot \vec{n} dS = 0$$

$$\oiint_{S_2} \vec{U} \cdot \vec{n} dS + \oiint_{S_3} \vec{U} \cdot \vec{n} dS = 0$$

➤ Continuity equation

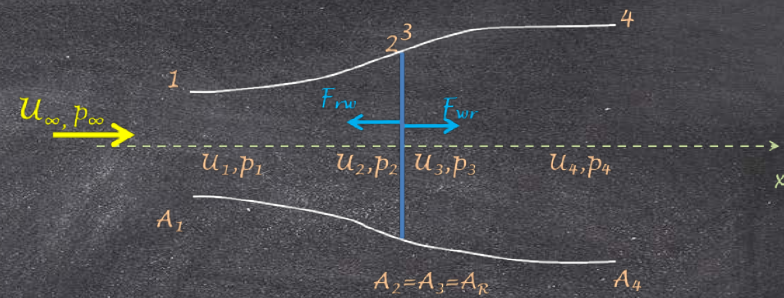
$$\oiint_S \vec{U} \cdot \vec{n} dS = 0$$

$$\oiint_{S_1} \vec{U} \cdot \vec{n} dS + \oiint_{S_t} \vec{U} \cdot \vec{n} dS + \oiint_{S_4} \vec{U} \cdot \vec{n} dS = 0$$

$$\oiint_{S_1} \vec{U} \cdot \vec{n} dS + \oiint_{S_{t1}} \vec{U} \cdot \vec{n} dS + \oiint_{S_2} \vec{U} \cdot \vec{n} dS = 0$$

$$\oiint_{S_3} \vec{U} \cdot \vec{n} dS + \oiint_{S_{t2}} \vec{U} \cdot \vec{n} dS + \oiint_{S_4} \vec{U} \cdot \vec{n} dS = 0$$

$$\oiint_{S_2} \vec{U} \cdot \vec{n} dS + \oiint_{S_3} \vec{U} \cdot \vec{n} dS = 0$$



➤ Continuity equation

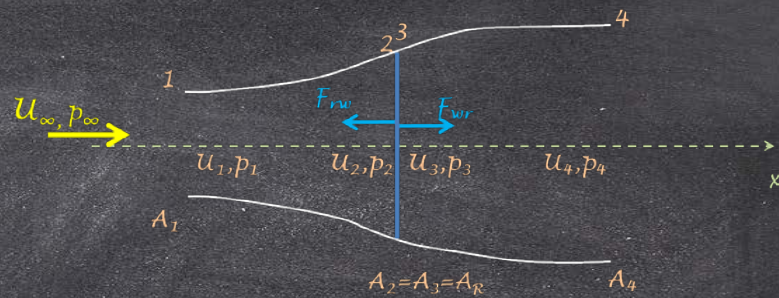
$$\oiint_S \vec{U} \cdot \vec{n} dS = 0$$

$$\oiint_{S_1} \vec{U} \cdot \vec{n} dS + \oiint_{S_4} \vec{U} \cdot \vec{n} dS = U_1 A_1 - U_4 A_4 = 0 \rightarrow U_1 = \frac{U_4 A_4}{A_1}$$

$$\oiint_{S_1} \vec{U} \cdot \vec{n} dS + \oiint_{S_2} \vec{U} \cdot \vec{n} dS = U_1 A_1 - U_2 A_2 = 0 \rightarrow U_1 = \frac{U_2 A_2}{A_1}$$

$$\oiint_{S_3} \vec{U} \cdot \vec{n} dS + \oiint_{S_4} \vec{U} \cdot \vec{n} dS = U_3 A_3 - U_4 A_4 = 0 \rightarrow U_3 = \frac{U_4 A_4}{A_3}$$

$$\oiint_{S_2} \vec{U} \cdot \vec{n} dS + \oiint_{S_3} \vec{U} \cdot \vec{n} dS = U_2 A_2 - U_3 A_3 = 0 \rightarrow U_2 = U_3$$



We assume U is constant in radial direction (one-dimensional model)

Setting the momentum equation

- We will define the momentum equation
- If you want to try to do it by yourself, pause the video now.



➤ Momentum equation in x-direction

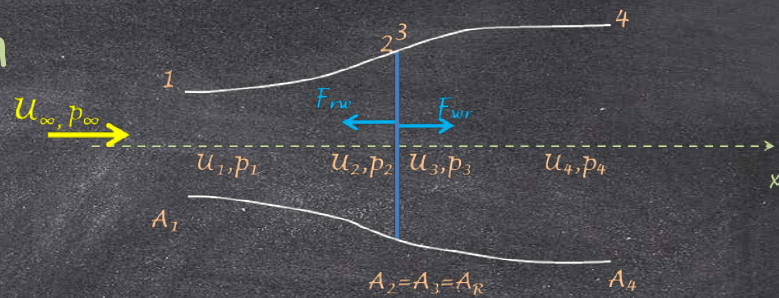
$$\oiint_S \rho \vec{U} (\vec{U} \cdot \vec{n} dS) = - \oiint_S p \vec{n} dS + \iiint_V \rho \vec{f} dV$$

$$\rho U_1^2 A_1 - \rho U_4^2 A_4 = -p_1 A_1 + p_4 A_4 - \oint_{S_t} p \vec{n} dS - F_{rw}$$

We assume p is constant
in radial direction
(one-dimensional model)

➤ Momentum equation in x-direction

$$\oiint_S \rho \vec{U} (\vec{U} \cdot \vec{n} dS) = - \oiint_S p \vec{n} dS + \iiint_V \rho \vec{f} dV$$



$$\rho U_1^2 A_1 - \rho U_4^2 A_4 = -p_1 A_1 + p_4 A_4 - \oiint_{S_t} p \vec{n} dS - F_{rw}$$

=0

We assume $p_1 = p_4 = p_\infty$ and that the total contribution of the pressure is null



$$\rho U_1^2 A_1 - \rho U_4^2 A_4 = -F_{rw}$$

➤ Momentum equation in x-direction

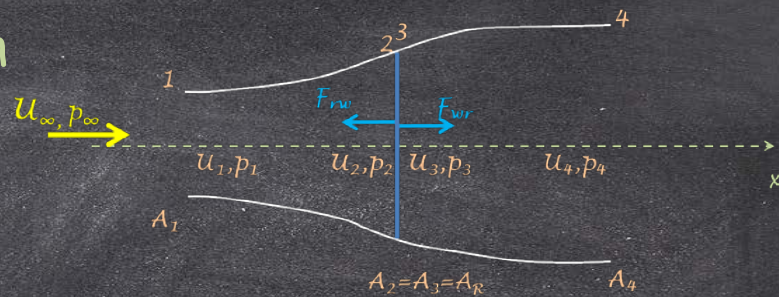
$$\oiint_S \rho \vec{U} (\vec{U} \cdot \vec{n} dS) = - \oiint_S p \vec{n} dS + \iiint_V \rho \vec{f} dV$$

$$\rho U_2^2 A_2 - \rho U_3^2 A_3 = -p_2 A_2 + p_3 A_3 - F_{rw}$$

=0



$$(p_2 - p_3) A_R = -F_{rw}$$



Setting Bernoulli's equation

- We will define Bernoulli's equation
- If you want to try to do it by yourself, pause the video now.

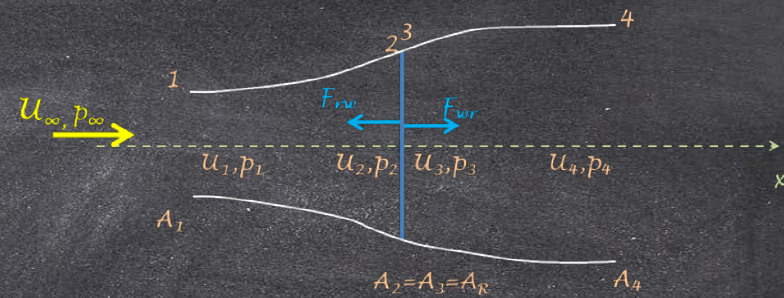


➤ Bernoulli's equation

$$p + \rho \frac{U^2}{2} = \text{constant}$$

$$p_1 + \frac{1}{2} \rho U_1^2 = p_2 + \frac{1}{2} \rho U_2^2$$

$$p_3 + \frac{1}{2} \rho U_3^2 = p_4 + \frac{1}{2} \rho U_4^2$$



Define equations for force and power and velocity at the disk

- If you want to try to do it by yourself,
pause the video now.

➤ Velocity at the actuator

$$U_R = \frac{1}{2}(U_1 + U_4)$$

➤ Thrust at the actuator

$$F_{wr} = \rho U_1^2 A_1 - \rho U_4^2 A_4 = (p_2 - p_3) A_R = \frac{1}{2} (\rho U_1^2 - \rho U_4^2) A_R$$

$$F_{wr} = \rho(U_1 - U_4)U_RA_R = 2\rho(U_1 - U_R)U_RA_R$$

➤ Power at the actuator

$$P = F_{wr} U_R = 2\rho(U_1 - U_R)U_R^2 A_R$$

What we have achieved:

- We have derived the actuator disk model
- We now have a model to determine power and/or force and/or the velocity in the rotor and/or the velocity in the far wake, given that we know the freestream conditions and at least one of these values.



Rotor and Wake Aerodynamics

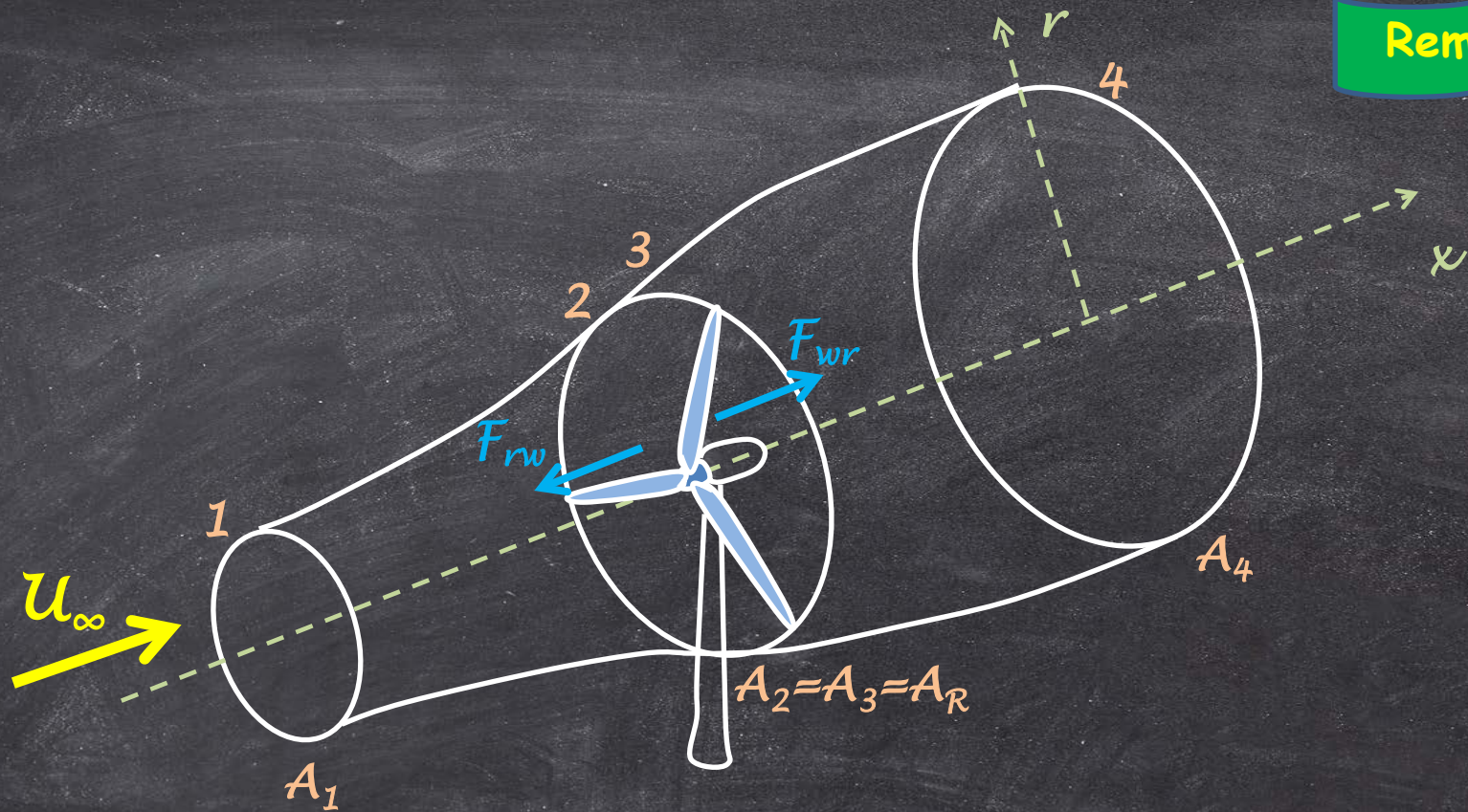
Performance of an Axial Actuator Disk

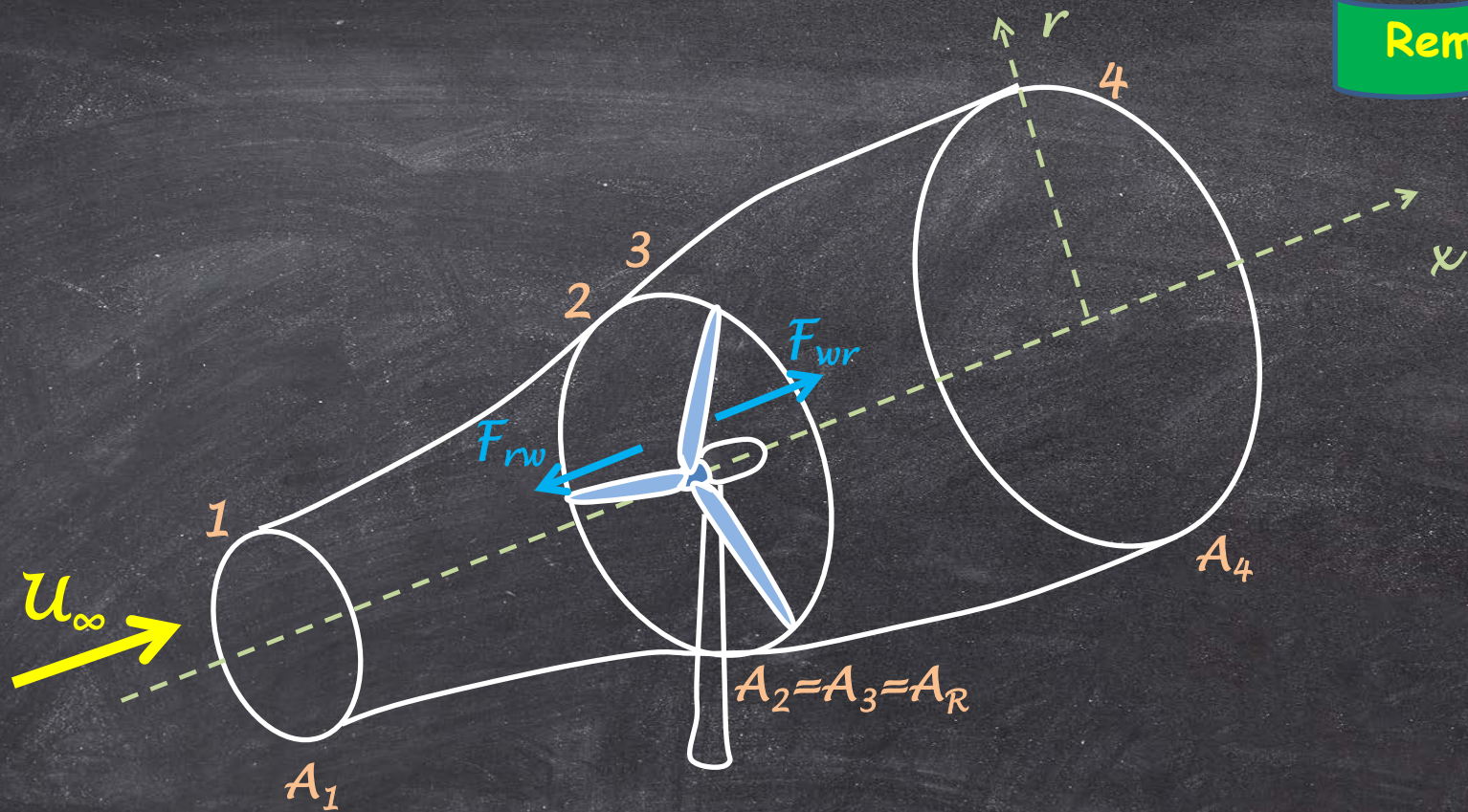


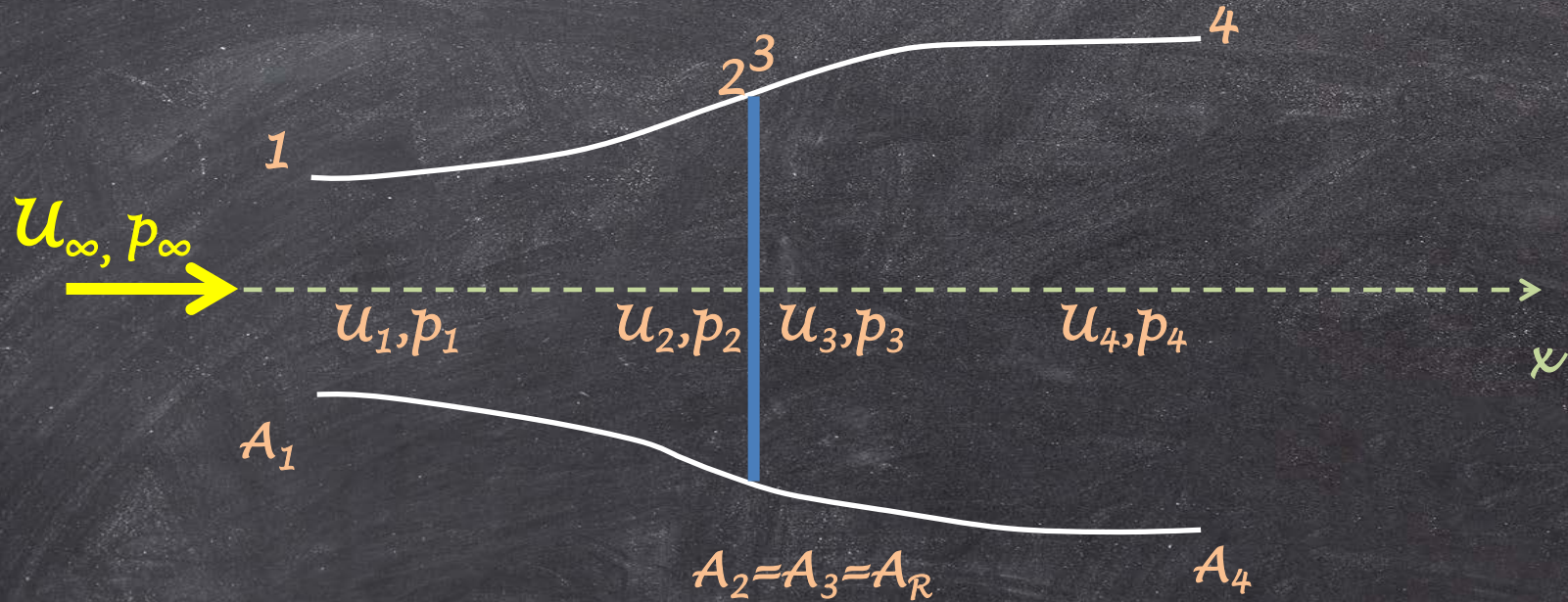
In the last module:

- We derived the actuator disk model
- We now have a model to determine power and/or force and/or the velocity in the rotor and/or the velocity in the far wake, given that we know the freestream conditions and at least one of these values.









➤ Continuity equation

$$\frac{\partial}{\partial t} \iiint_V \rho dV + \iint_S \rho \vec{U} \cdot \vec{n} dS = 0$$

➤ Momentum equation

$$\frac{\partial}{\partial t} \iiint_V \rho \vec{U} dV + \iint_S \rho \vec{U} (\vec{U} \cdot \vec{n} dS) = - \iint_S p \vec{n} dS + \iiint_V \rho \vec{f} dV + F_{viscous}$$

➤ Energy equation

$$\frac{\partial}{\partial t} \iiint_V \left(\rho e + \rho \frac{U^2}{2} \right) dV + \iint_S \left(\rho e + \rho \frac{U^2}{2} \right) (\vec{U} \cdot \vec{n} dS) =$$

$$\frac{\partial}{\partial t} \iiint_V \dot{q} dV + \dot{Q}_{viscous} - \iint_S p (\vec{U} \cdot \vec{n} dS) + \iiint_V \rho \vec{f} \cdot \vec{U} dV + \dot{W}_{viscous}$$

➤ Velocity at the actuator

$$U_R = \frac{1}{2}(U_1 + U_4)$$

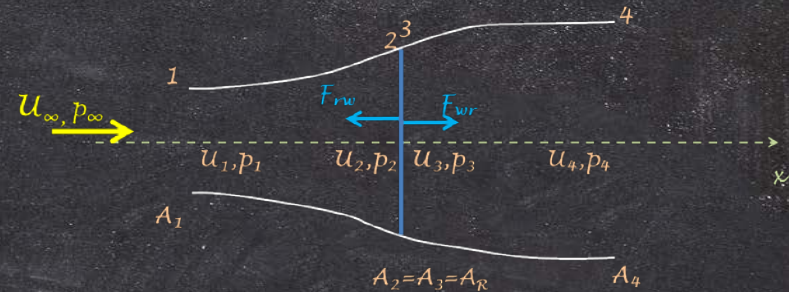
➤ Thrust at the actuator

$$F_{rw} = \rho U_1^2 A_1 - \rho U_4^2 A_4 = (p_2 - p_3) A_R = \frac{1}{2}(\rho U_1^2 - \rho U_4^2) A_R$$

$$F_{rw} = \rho(U_1 - U_4)U_R A_R = 2\rho(U_1 - U_R)U_R A_R$$

➤ Power at the actuator

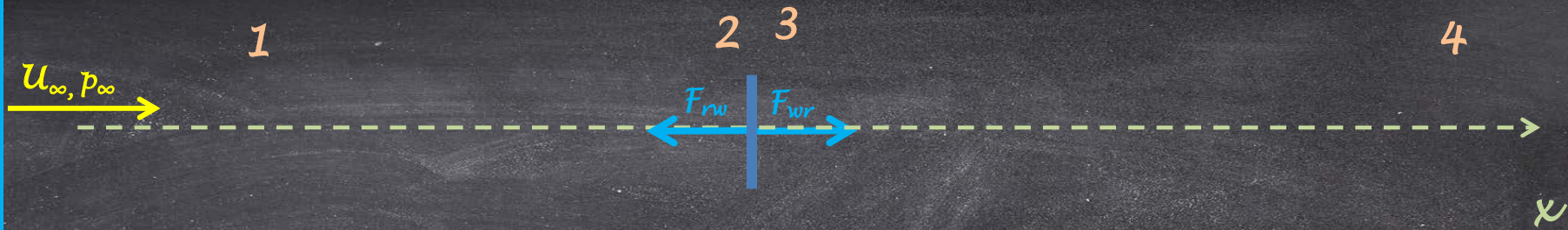
$$P = F_{rw} U_R = 2\rho(U_1 - U_R)U_R^2 A_R$$

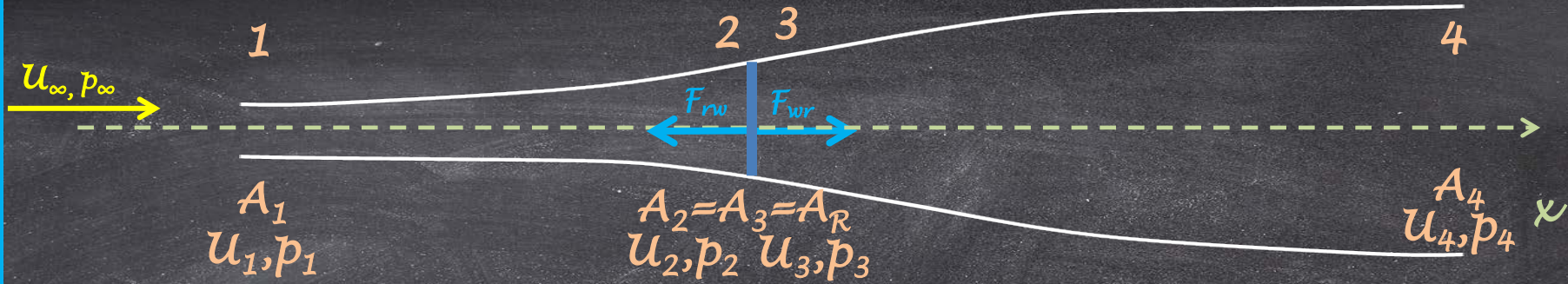


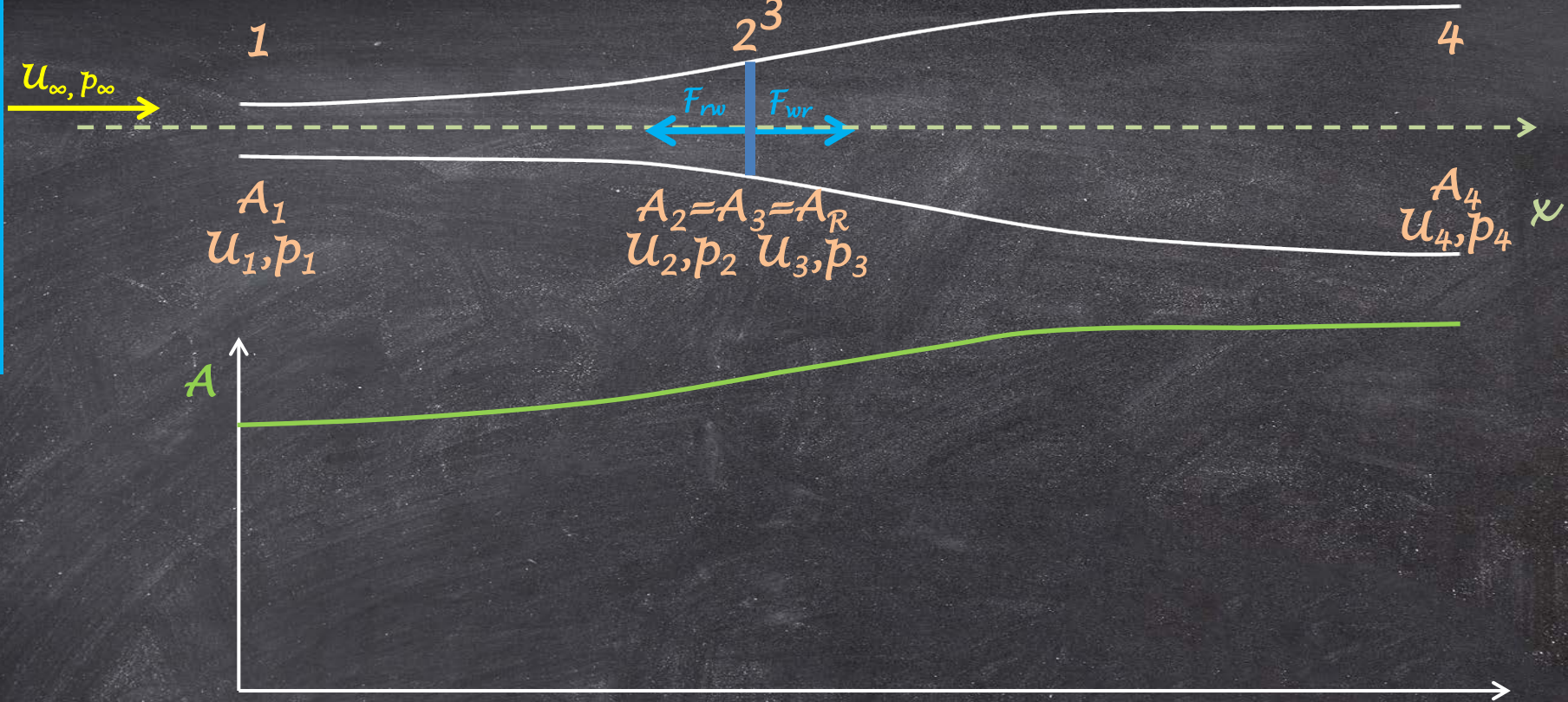
Our objective for this lecture

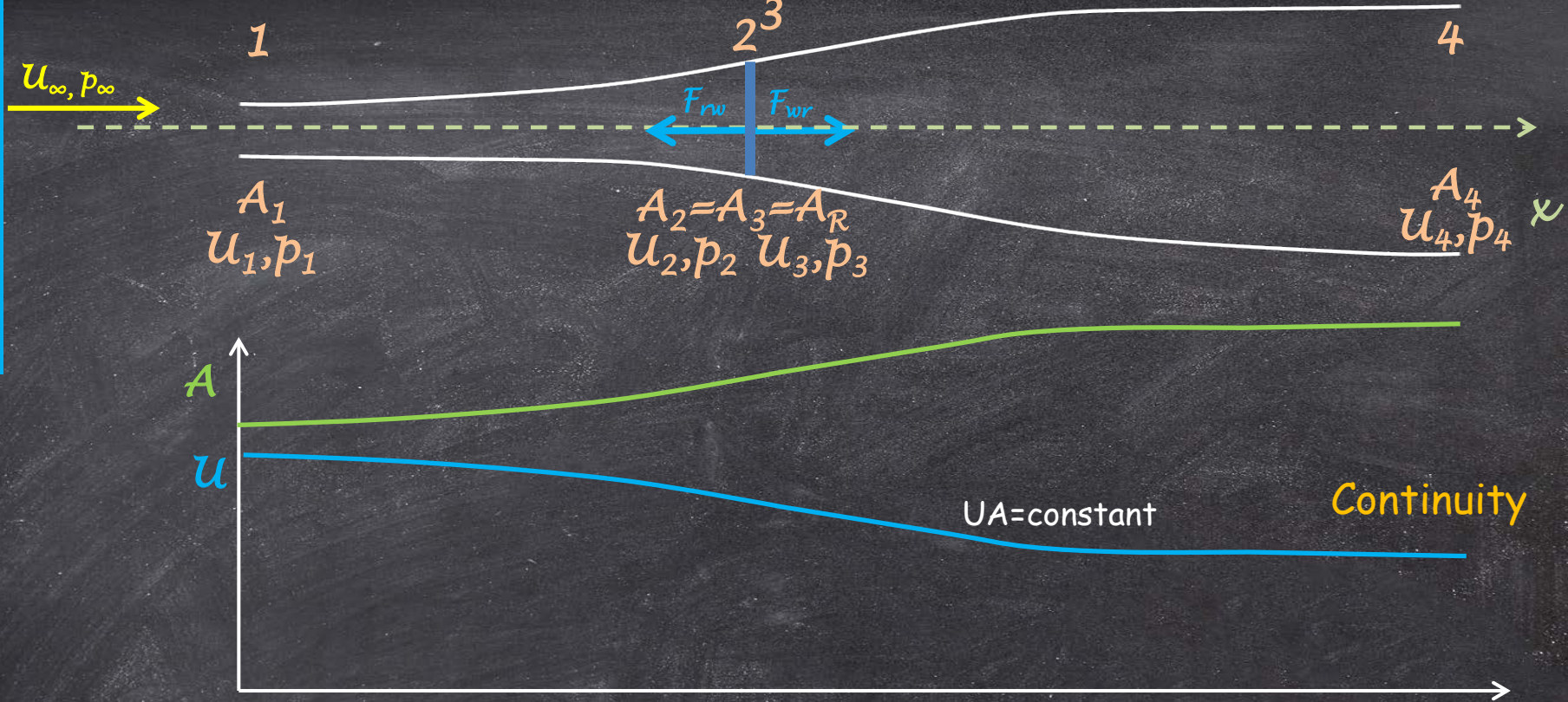
- Evaluate the performance of an axial actuator disk
 - Flow properties, loads and power, applied to wind turbine and propeller modes
 - Limits of the model

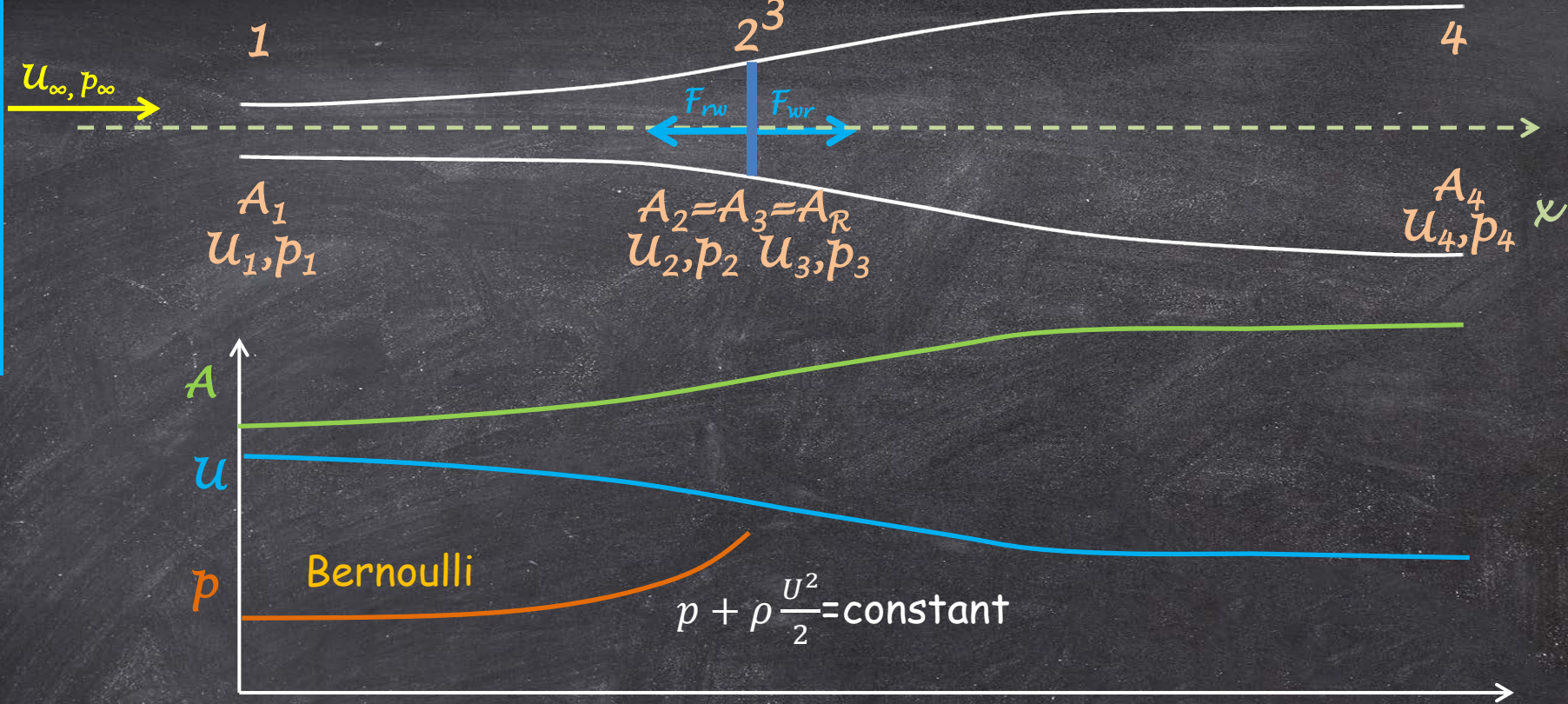


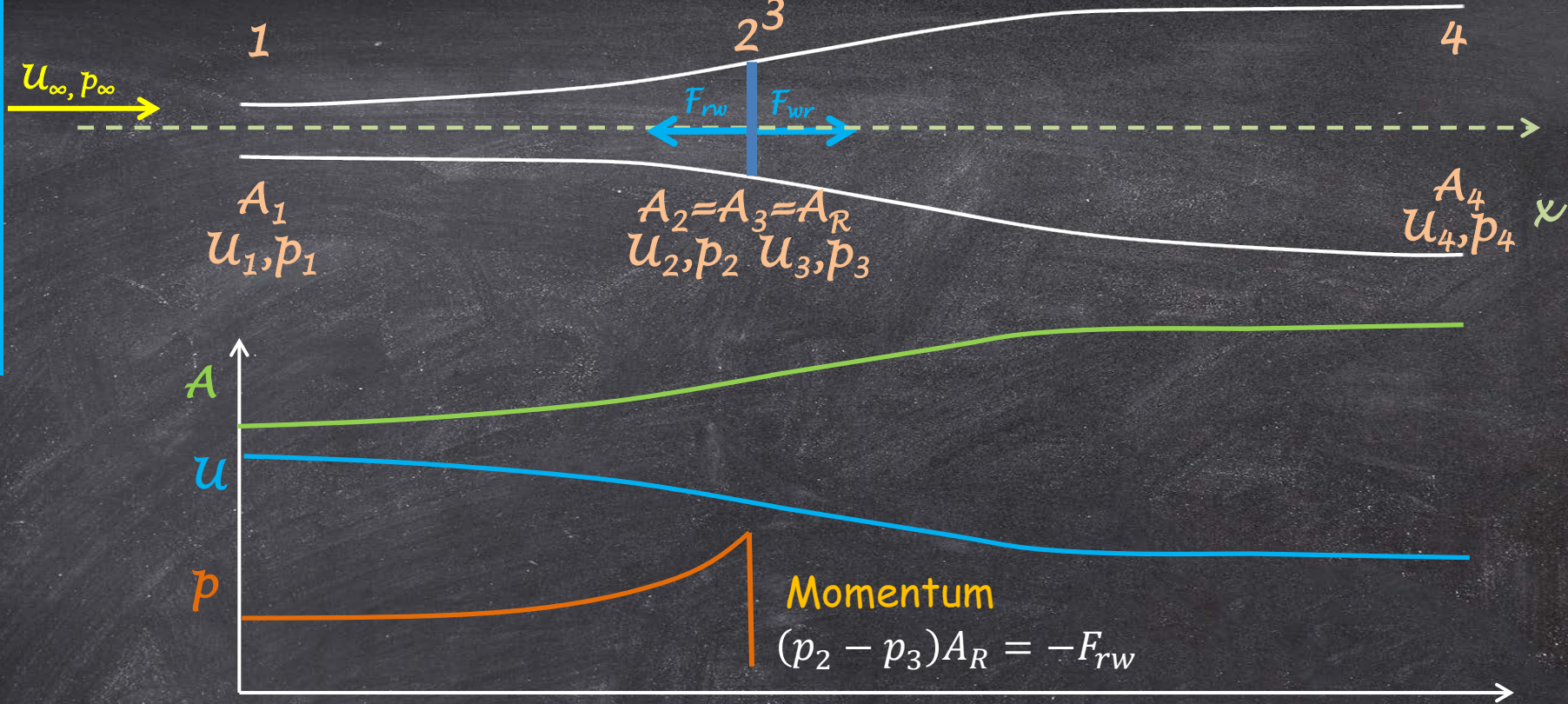


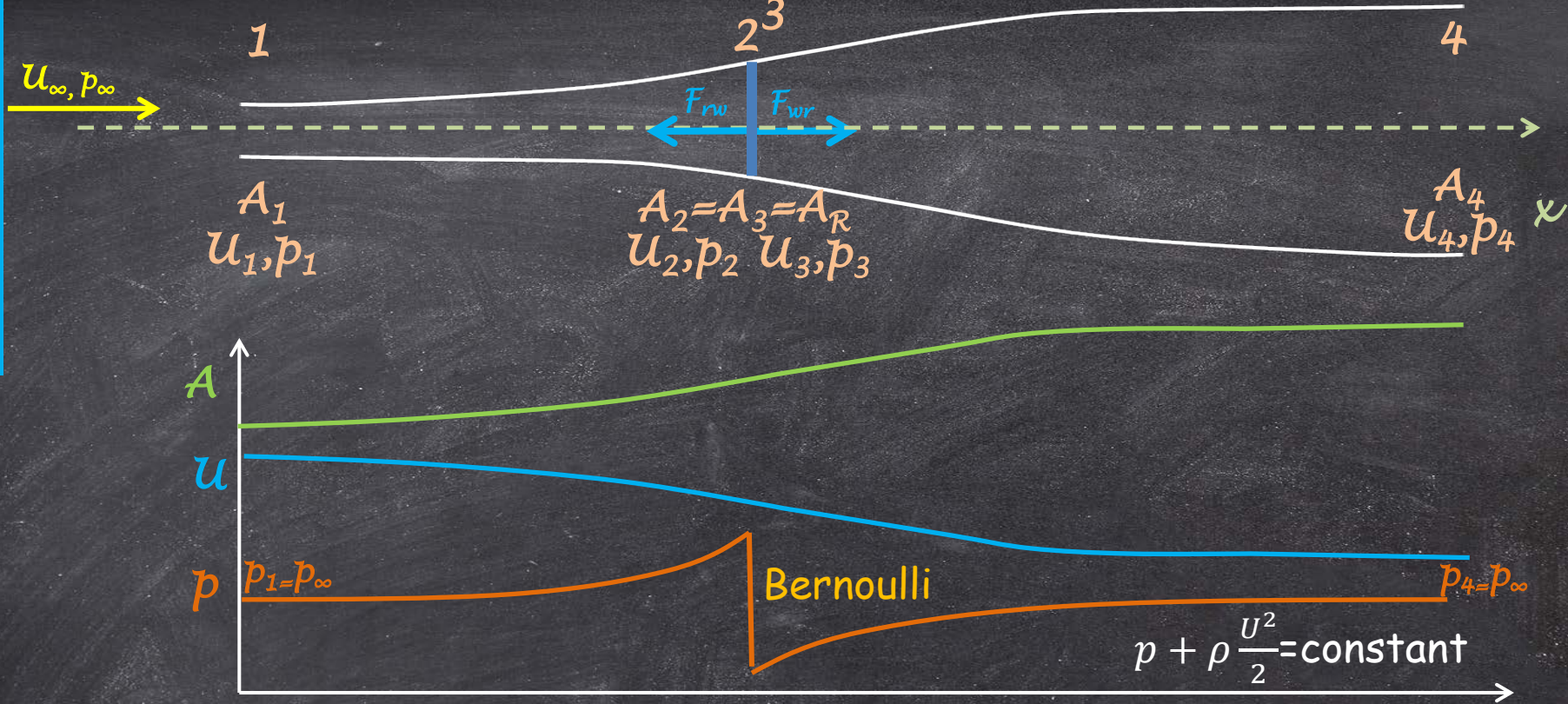


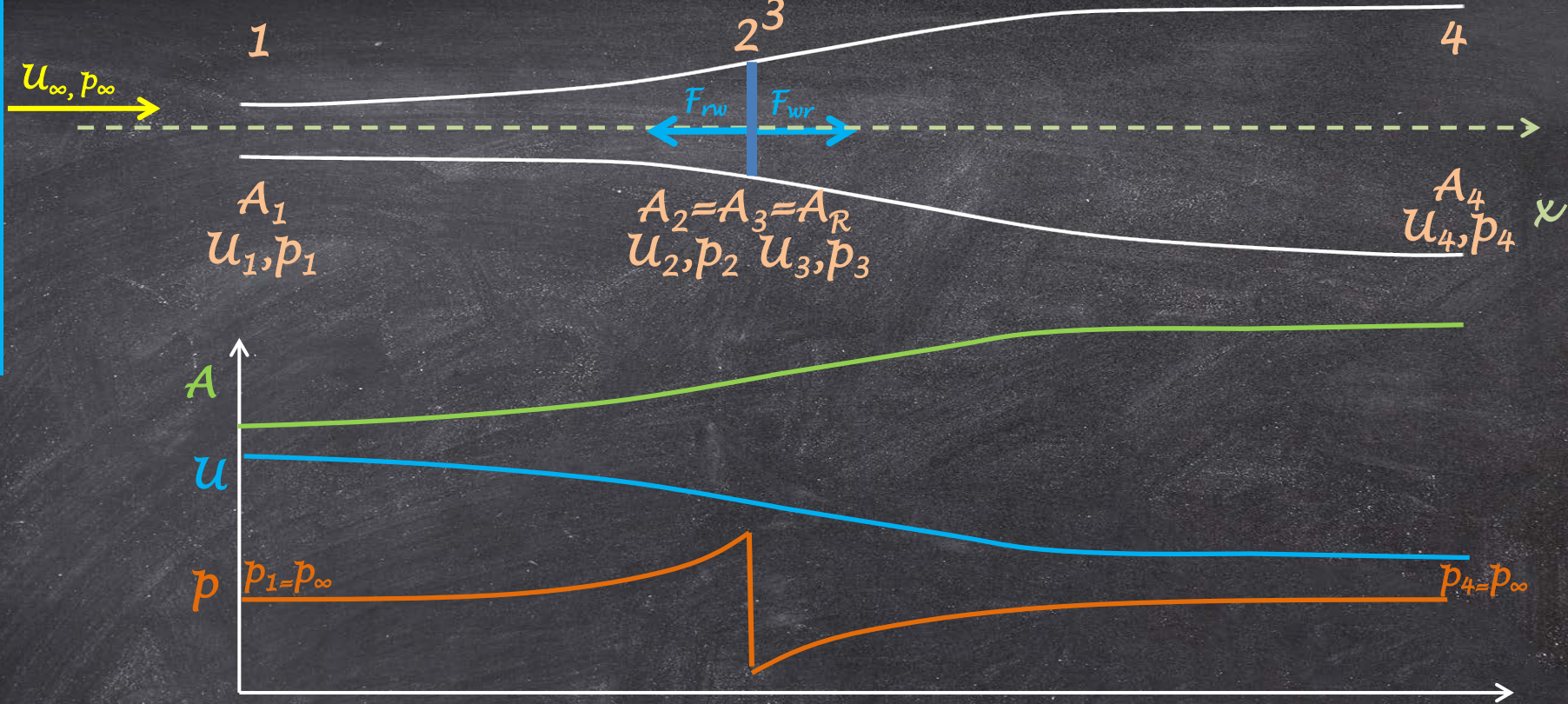


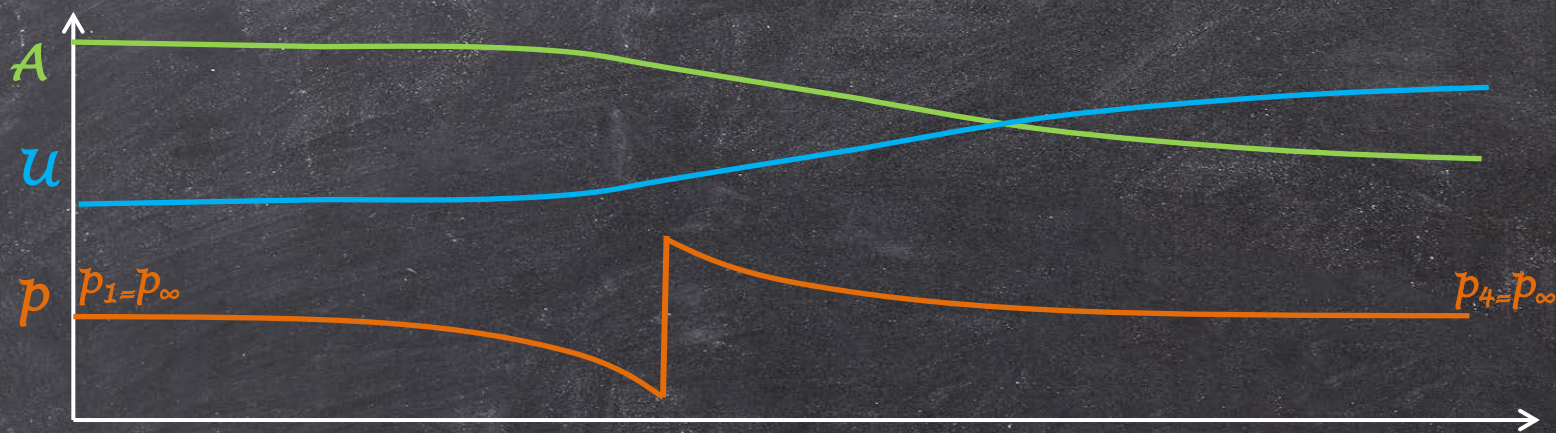
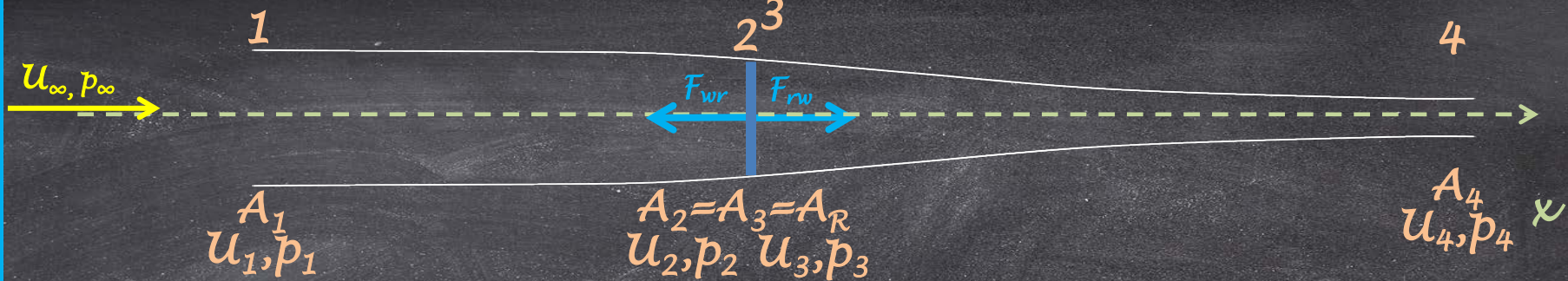












Limits of the actuator disk model

- When we derived the actuator disk model, we stated that the problem is one-dimensional (U and p are constant in radial direction)
- What does this imply in terms of validity of the model?



Rotor in axial flow

- Consider the relations between the velocity of motion of the rotor, the velocity of the wind and the loading of the rotor.
- How many different configurations are possible?



Introducing induction factor

- The induction factor a determines the relation between the velocity at the disk and the unperturbed velocity

$$U_R = (1 - a)U_\infty$$

- Note: in propeller books, usually defined as $U_R = (1 + a)U_\infty$



➤ Velocity at the actuator

$$U_R = \frac{1}{2}(U_1 + U_4)$$

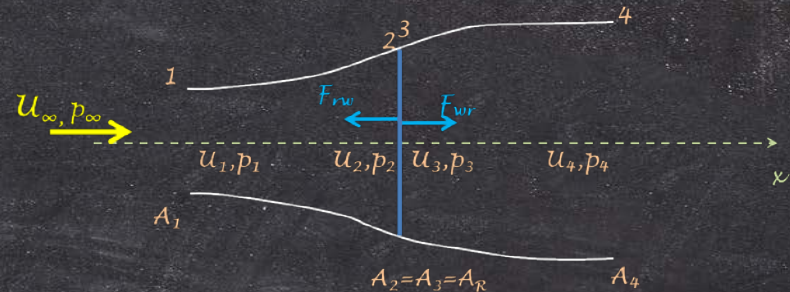
➤ Thrust at the actuator

$$F_{rw} = \rho U_1^2 A_1 - \rho U_4^2 A_4 = (p_2 - p_3) A_R = \frac{1}{2}(\rho U_1^2 - \rho U_4^2) A_R$$

$$F_{rw} = \rho(U_1 - U_4)U_R A_R = 2\rho(U_1 - U_R)U_R A_R$$

➤ Power at the actuator

$$P = F_{rw} U_R = 2\rho(U_1 - U_R)U_R^2 A_R$$



Re-writing the equations

➤ Velocity at the actuator

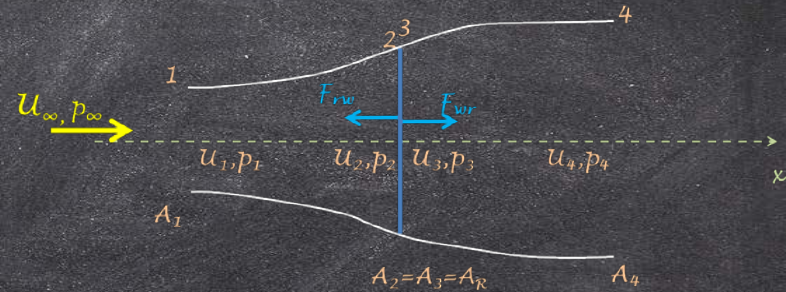
$$\frac{U_R}{U_\infty} = (1 - a)$$

➤ Thrust at the actuator

$$\frac{F_{rw}}{\frac{1}{2}\rho U_\infty^2 A_R} = C_T = 4a(1 - a)$$

➤ Power at the actuator

$$\frac{P}{\frac{1}{2}\rho U_\infty^3 A_R} = C_P = 4a(1 - a)^2$$

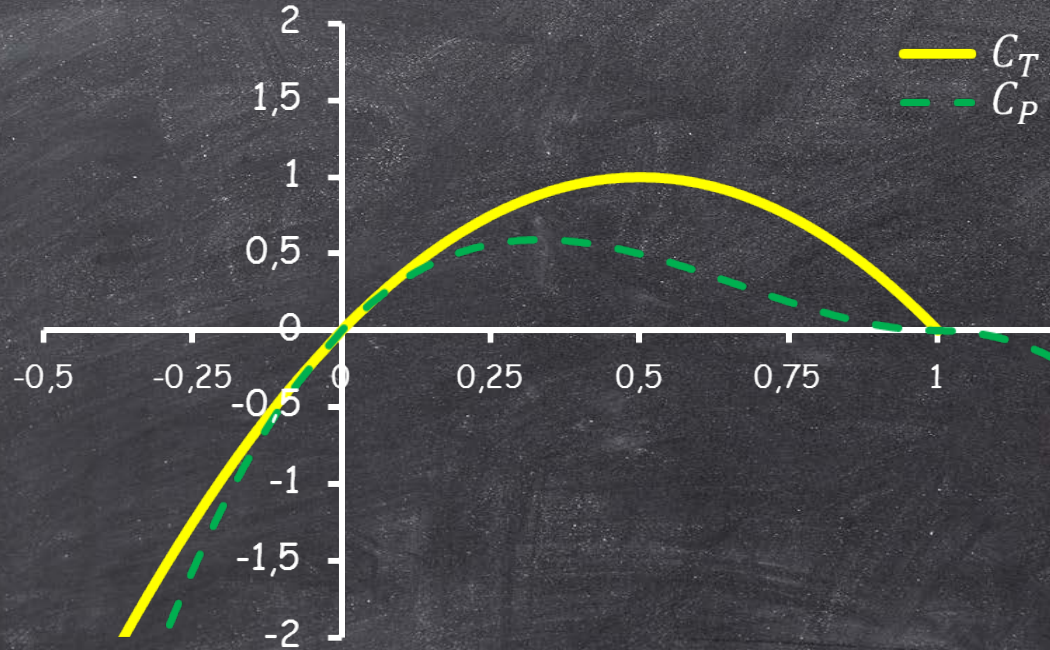


Thrust coefficient

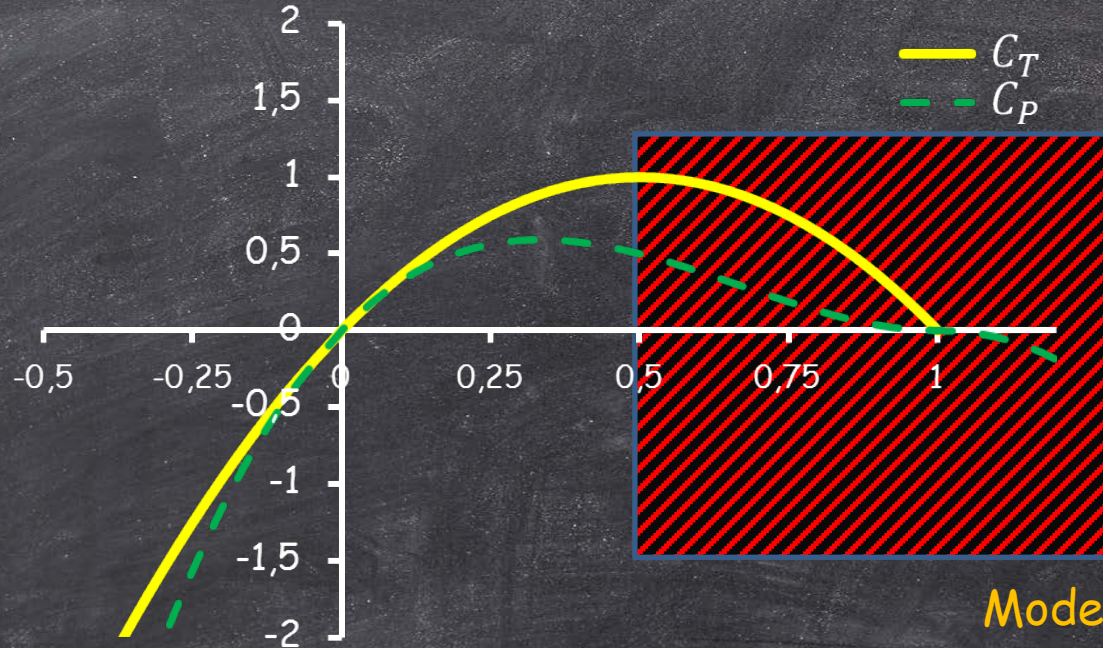
Power coefficient



C_T and C_P as a function of a



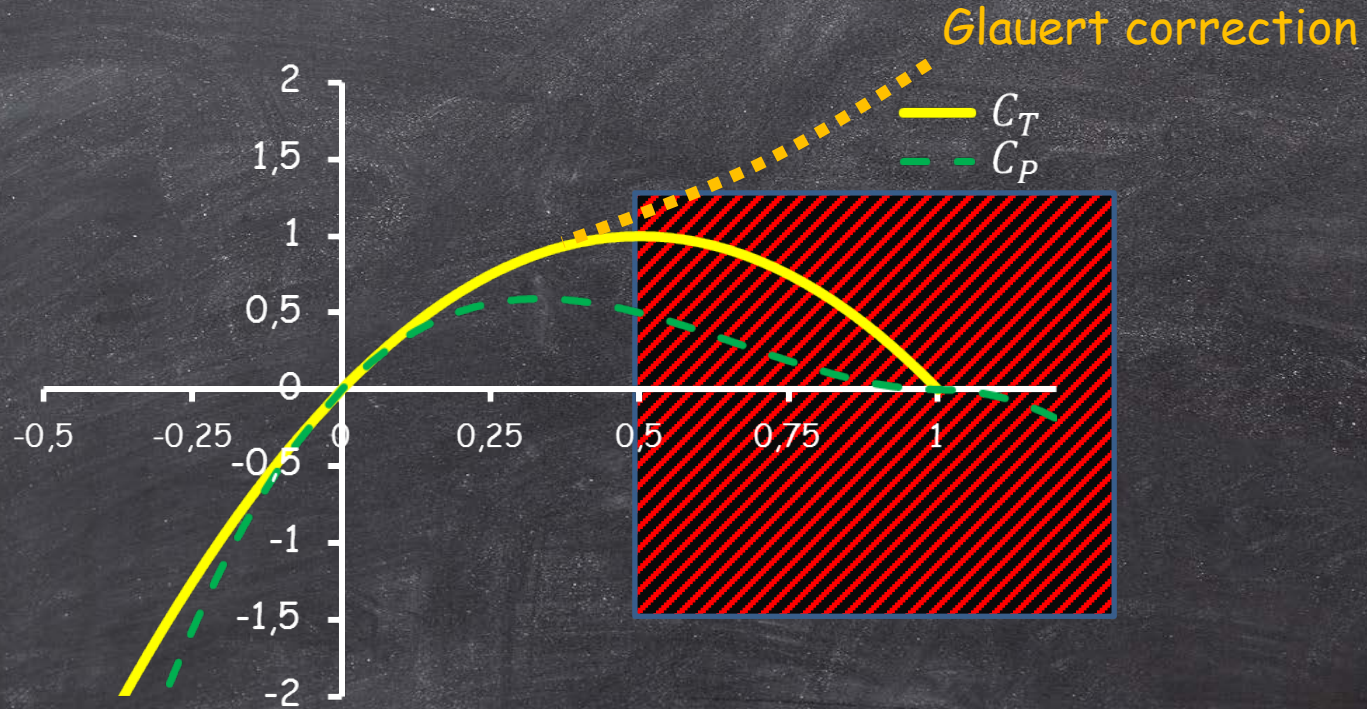
C_T and C_P as a function of a



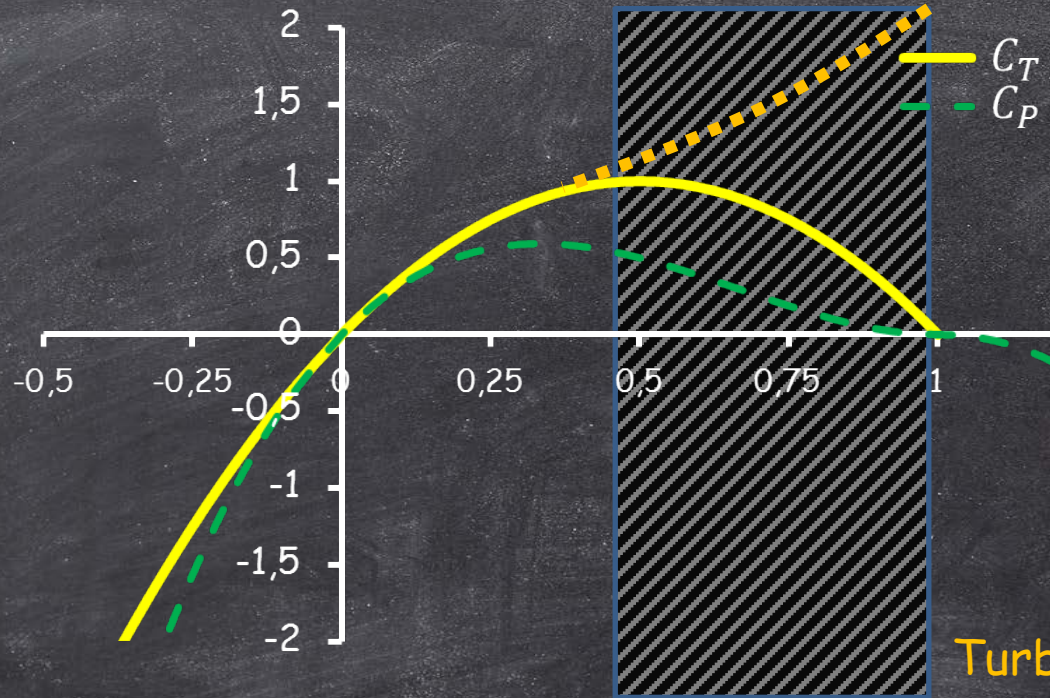
Model is invalid



C_T and C_P as a function of a



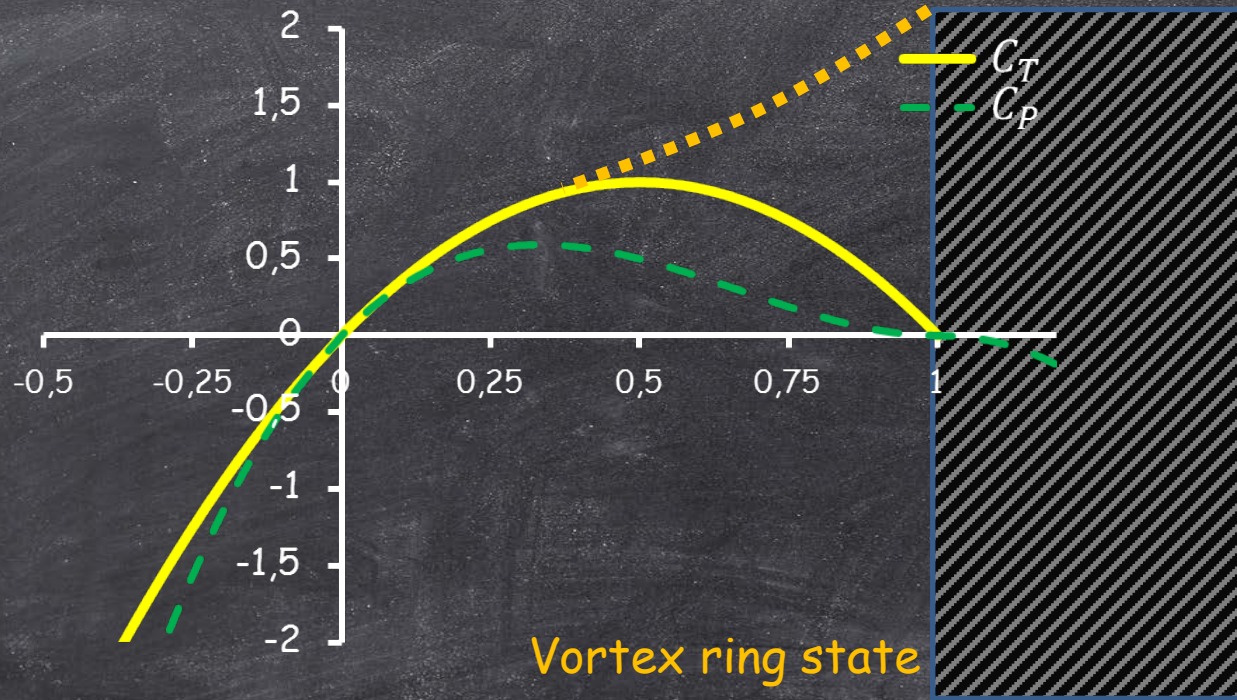
C_T and C_P as a function of a



Turbulent wake state



C_T and C_p as a function of a



Betz limit

What is the maximum power we can retrieve from a wind turbine?

Remember: $C_P = 4a(1 - a)^2$

The power curve maximum is found with $\frac{dC_P}{da} = 4(1 - a)(1 - 3a) = 0$



$$a = \frac{1}{3}$$

$$C_{P_{max}} = \frac{16}{27} = 0.593$$



In this module:

We derived the performance of an actuator disk model, based on the induction factor a

We derived the Betz limit for maximum power extracted from an actuator disk



Rotor and Wake Aerodynamics

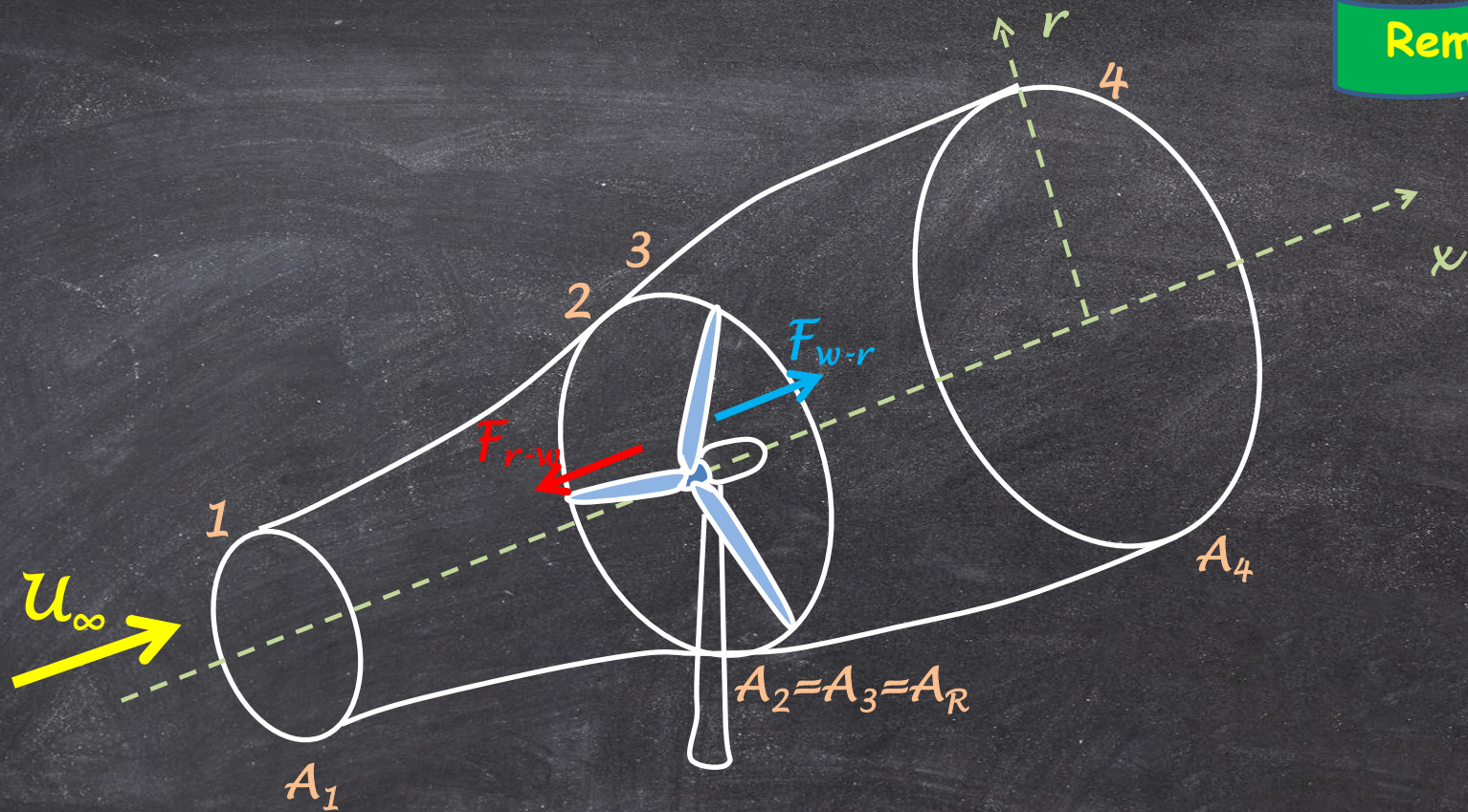
Blade Element Momentum Theory

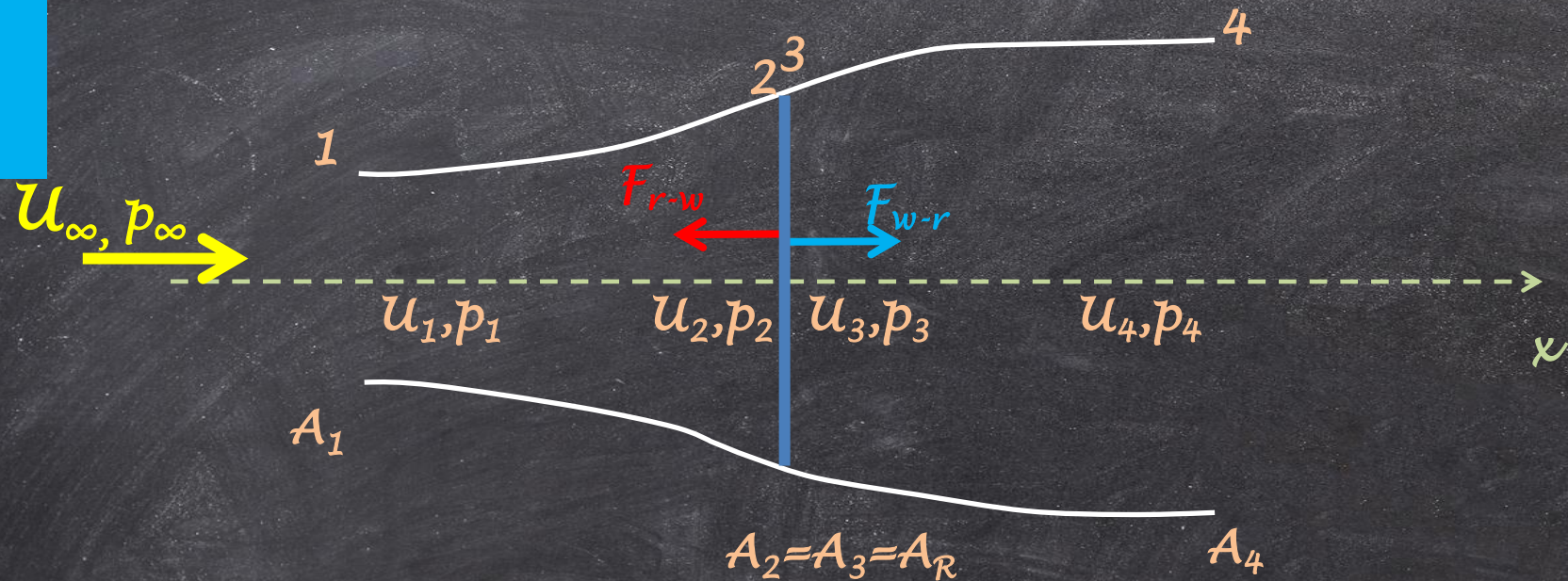
In the previous modules:

- We derived the actuator disk model
- We used the actuator disk model to calculate the performance of the rotor using a one-dimensional model
 - Pressure, velocity and loading are uniform at each section of the streamtube

Our objective for today:

- To derive the Blade Element Momentum Theory model (BEM)
 - Pressure, velocity and loading are no longer uniform over each section of the streamtube
 - We relate the loading in the streamtube to the loading on the section of the rotor's blade which crosses the streamtube





➤ Continuity equation

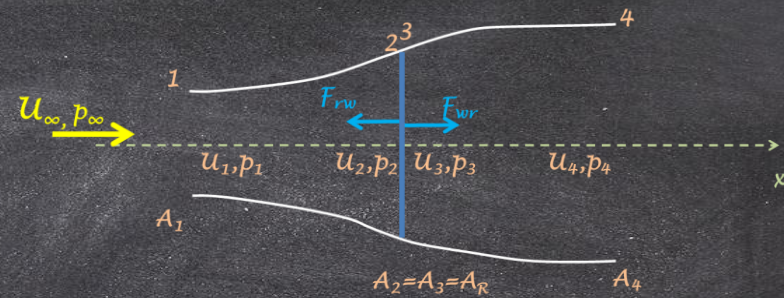
$$\oiint_S \vec{U} \cdot \vec{n} dS = 0$$

$$\oiint_{S_1} \vec{U} \cdot \vec{n} dS + \oiint_{S_4} \vec{U} \cdot \vec{n} dS = U_1 A_1 - U_4 A_4 = 0 \rightarrow U_1 = \frac{U_4 A_4}{A_1}$$

$$\oiint_{S_1} \vec{U} \cdot \vec{n} dS + \oiint_{S_2} \vec{U} \cdot \vec{n} dS = U_1 A_1 - U_2 A_2 = 0 \rightarrow U_1 = \frac{U_2 A_2}{A_1}$$

$$\oiint_{S_3} \vec{U} \cdot \vec{n} dS + \oiint_{S_4} \vec{U} \cdot \vec{n} dS = U_3 A_3 - U_4 A_4 = 0 \rightarrow U_3 = \frac{U_4 A_4}{A_3}$$

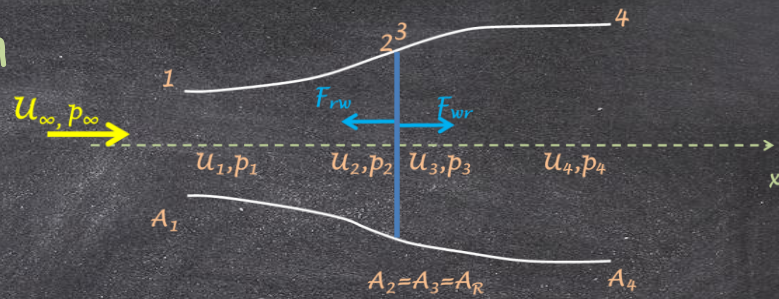
$$\oiint_{S_2} \vec{U} \cdot \vec{n} dS + \oiint_{S_3} \vec{U} \cdot \vec{n} dS = U_2 A_2 - U_3 A_3 = 0 \rightarrow U_2 = U_3$$



Reminder

➤ Momentum equation in x-direction

$$\oiint_S \rho \vec{U} (\vec{U} \cdot \vec{n} dS) = - \oiint_S p \vec{n} dS + \iiint_V \rho \vec{f} dV$$



$$\rho U_2^2 A_2 - \rho U_3^2 A_3 = -p_2 A_2 + p_3 A_3 - F_{rw}$$

$$(p_2 - p_3) A_R = -F_{rw}$$

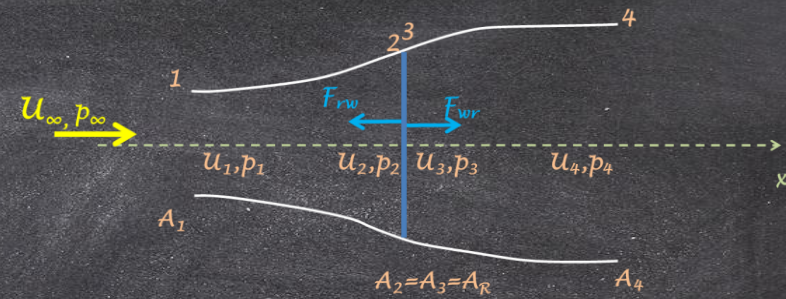
Reminder

➤ Bernoulli's equation

$$p + \rho \frac{U^2}{2} = \text{constant}$$

$$p_1 + \frac{1}{2} \rho U_1^2 = p_2 + \frac{1}{2} \rho U_2^2$$

$$p_3 + \frac{1}{2} \rho U_3^2 = p_4 + \frac{1}{2} \rho U_4^2$$



Reminder

➤ Velocity at the actuator

$$U_R = \frac{1}{2}(U_1 + U_4)$$

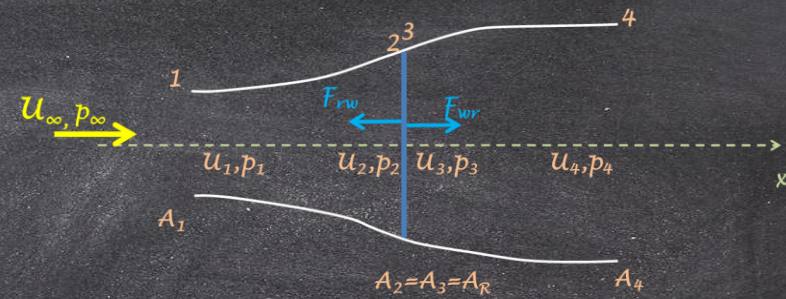
➤ Thrust at the actuator

$$F_{rw} = \rho U_1^2 A_1 - \rho U_4^2 A_4 = (p_2 - p_3) A_R = \frac{1}{2}(\rho U_1^2 - \rho U_4^2) A_R$$

$$F_{rw} = \rho(U_1 - U_4)U_R A_R = 2\rho(U_1 - U_R)U_R A_R$$

➤ Power at the actuator

$$P = F_{rw}U_R = 2\rho(U_1 - U_R)U_R^2 A_R$$



Reminder

Re-writting the equations

➤ Velocity at the actuator

$$\frac{U_R}{U_\infty} = (1 - a) \quad \frac{U_4}{U_\infty} = (1 - 2a)$$

➤ Thrust at the actuator

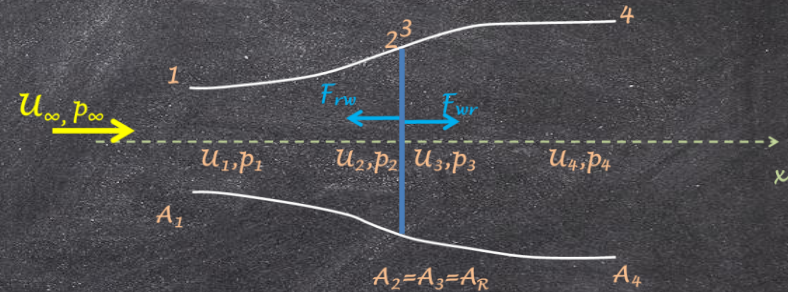
$$\frac{F_{rw}}{\frac{1}{2} \rho U_\infty^2 A_R} = C_T = 4a(1 - a)$$

Thrust coefficient

➤ Power at the actuator

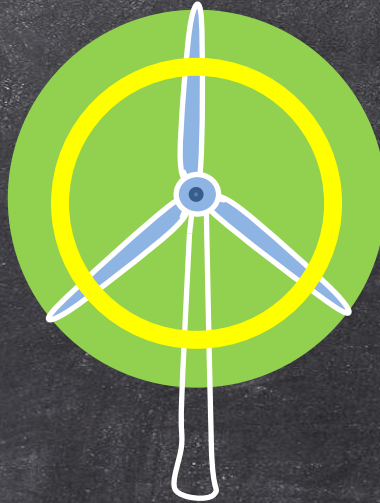
$$\frac{P}{\frac{1}{2} \rho U_\infty^3 A_R} = C_P = 4a(1 - a)^2$$

Power coefficient

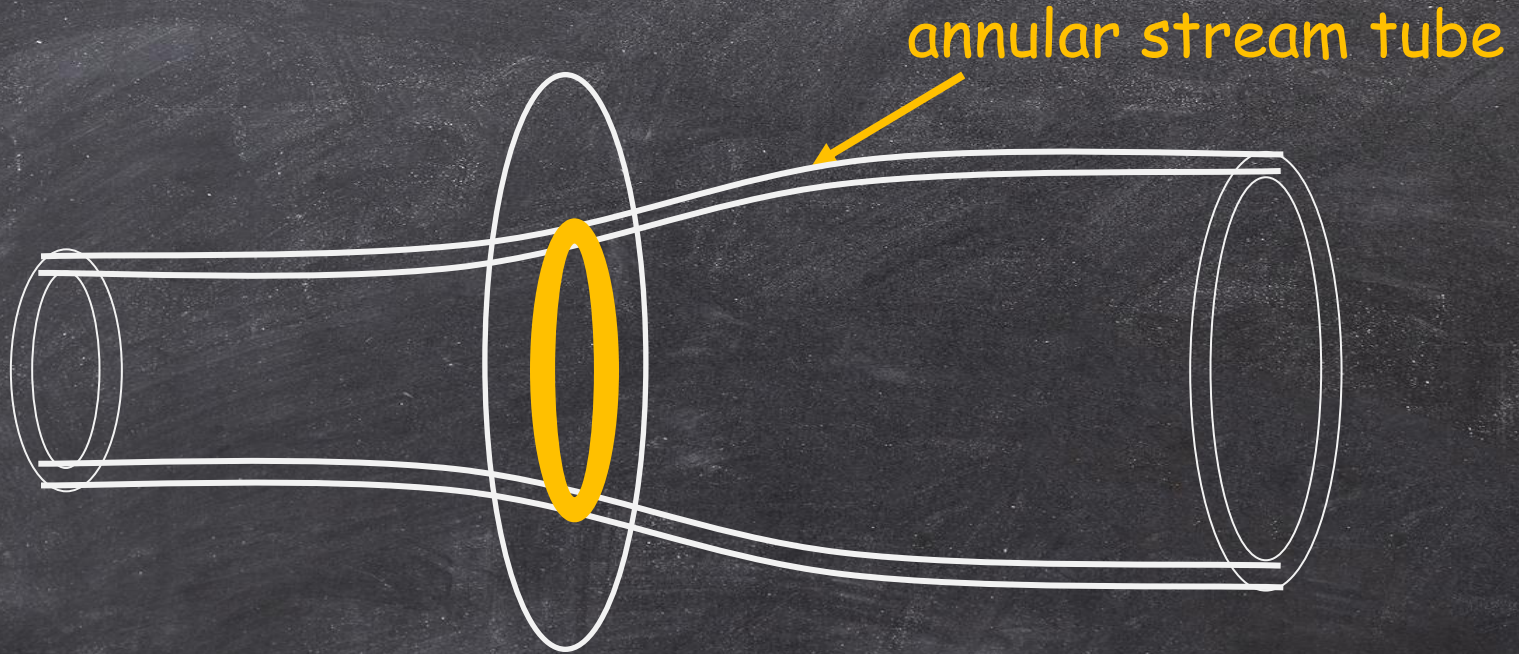


Reminder

We will now divide the disk in several annuli

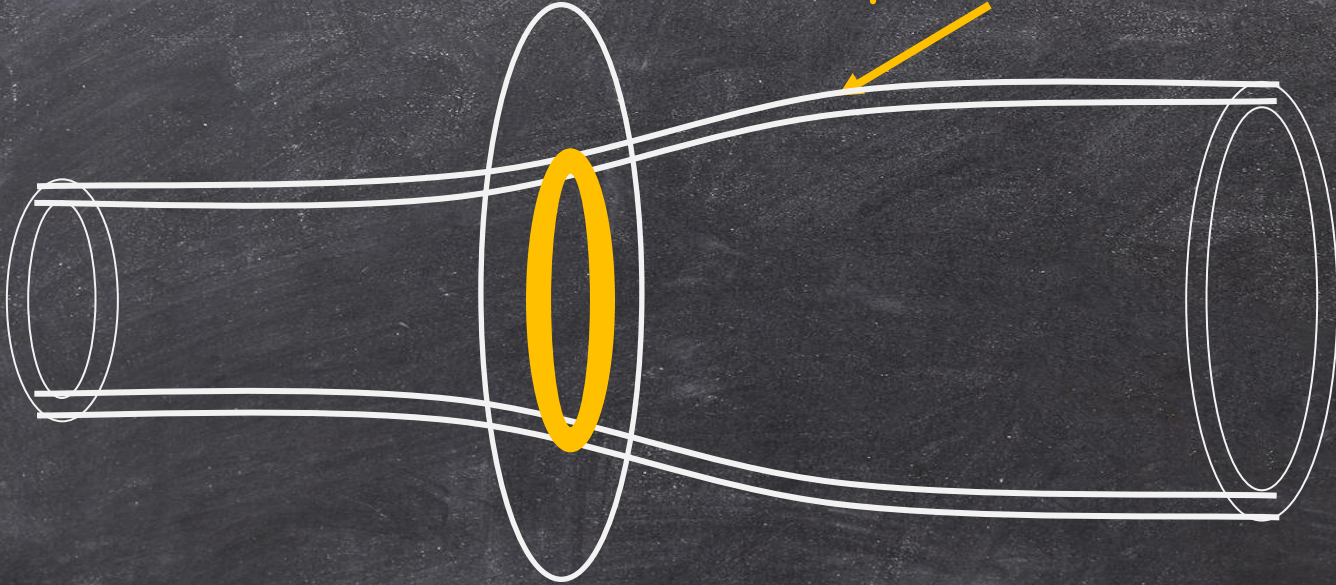


We apply momentum theory
to each annular streamtube



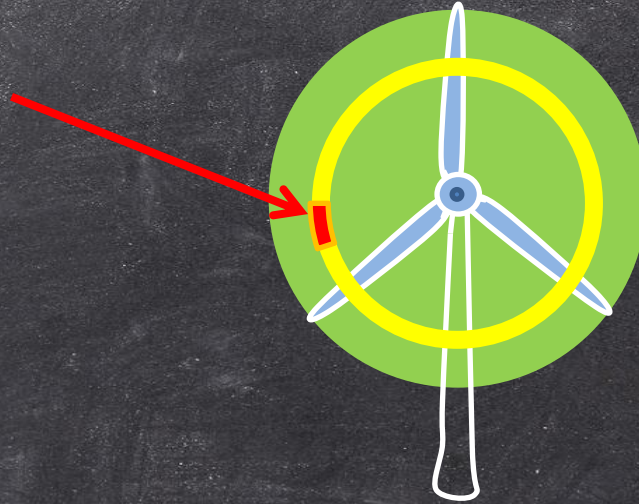
We apply momentum theory
to each annular streamtube

We assume independence of the annuli



We can discretize the annulus in azimuthal sections

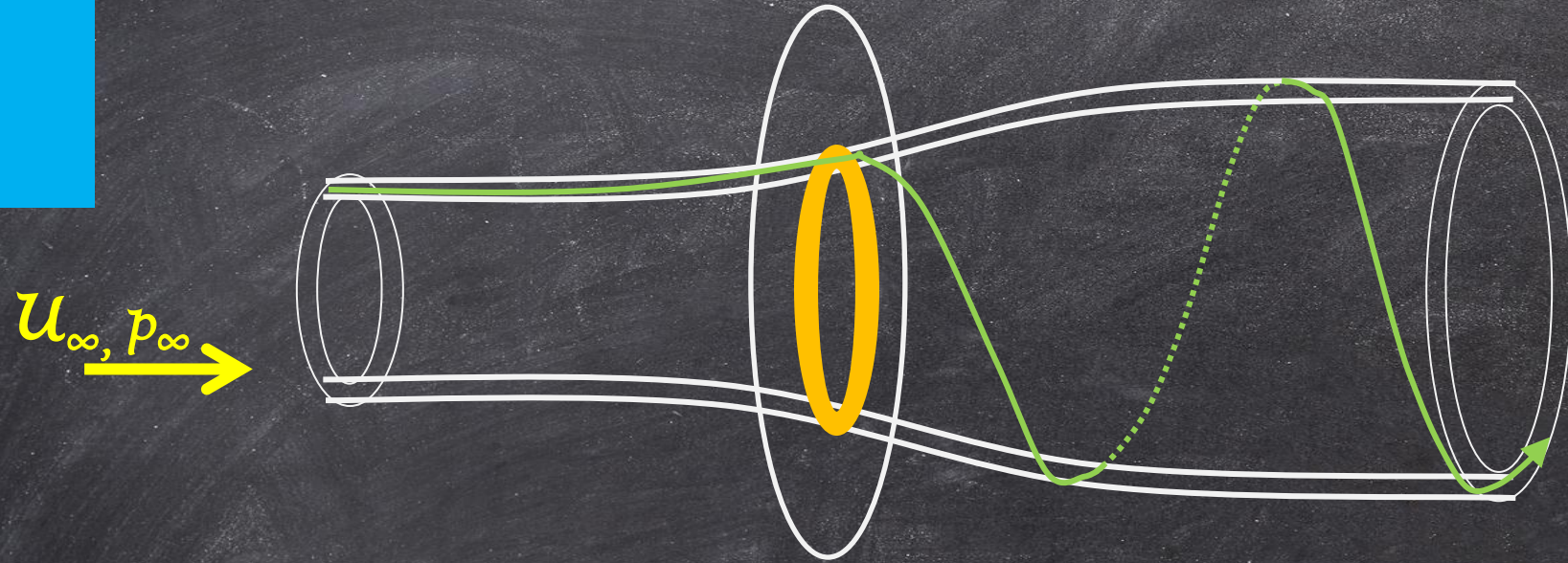
We then apply the same equations,
but with the area and loading
proportional to the radial and
azimuthal scale of the section



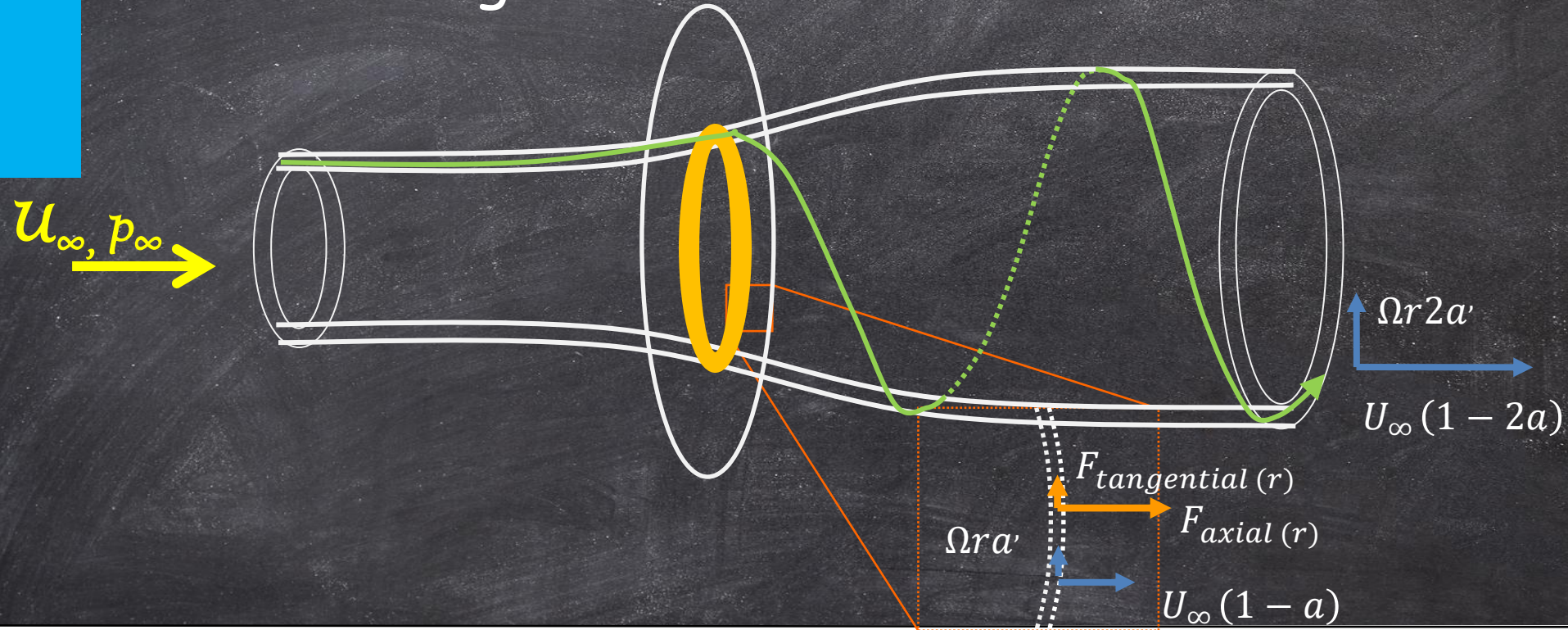
What loading is applied in this streamtube annulus?

- Axial loading, resulting in axial induction
 - Total axial loading will result in thrust
- Tangential loading, resulting in tangential induction
 - Total tangential loading will result in torque

Torque will create a rotation in the wake

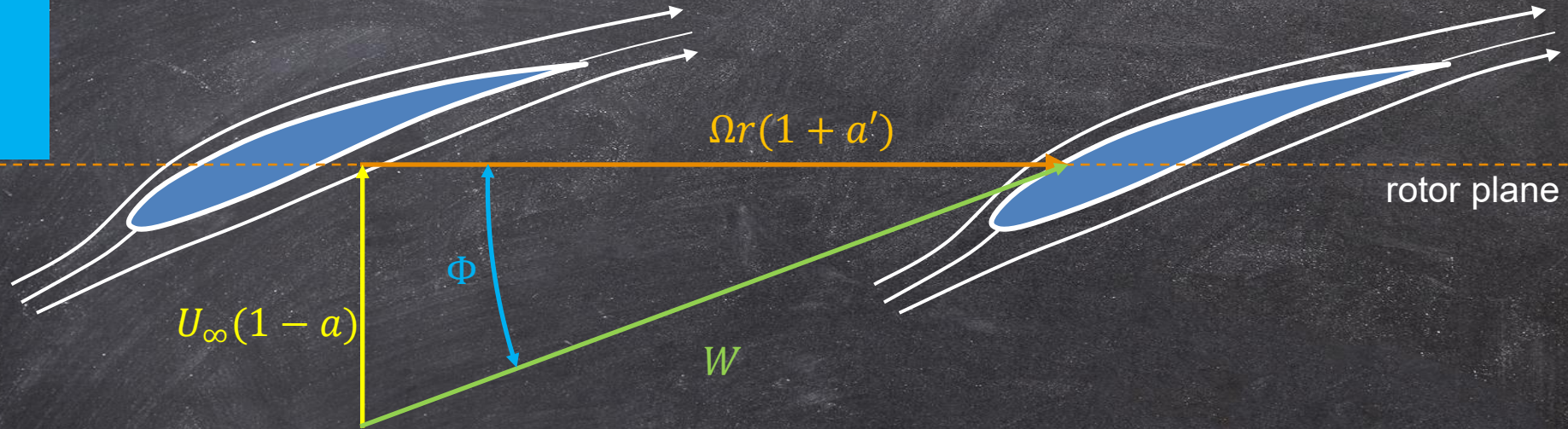


We have at each annulus
a tangential and axial induction



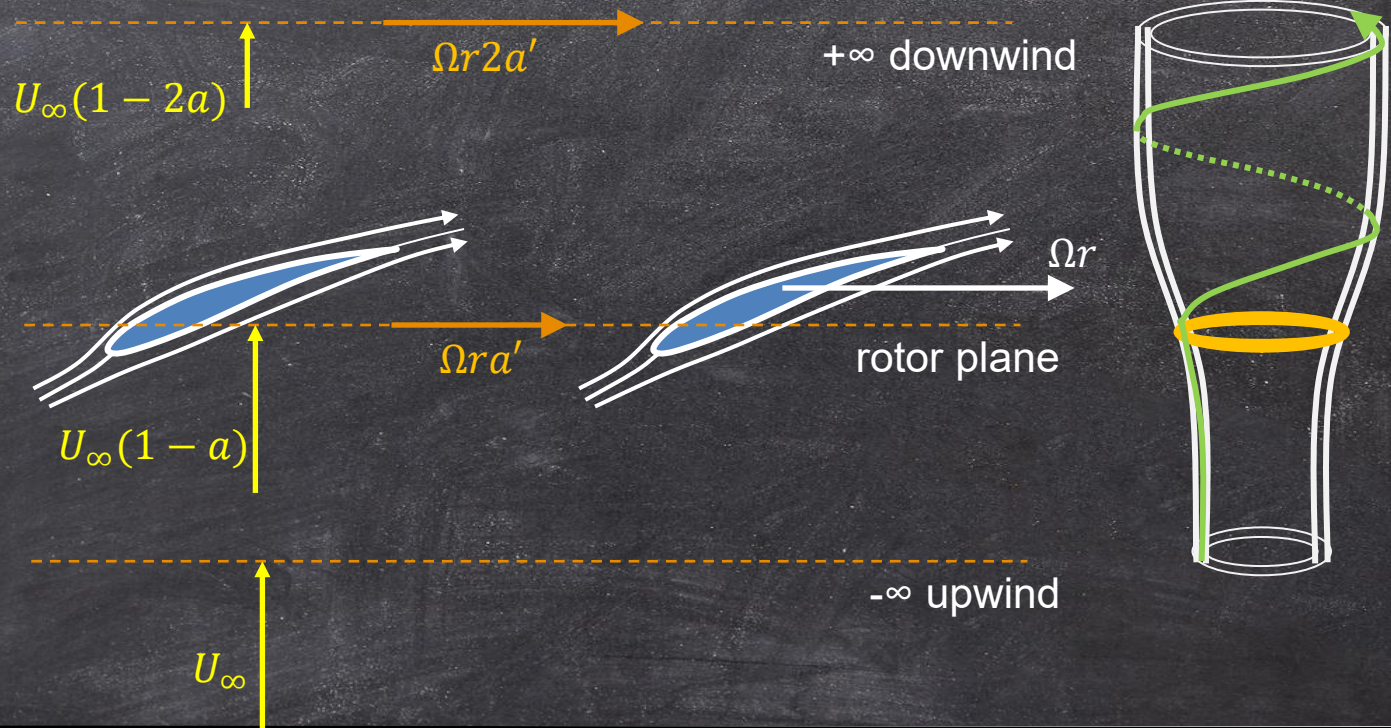
Wake rotation (wind turbine)

Rotating frame of reference



Wake rotation (wind turbine)

Inertial frame of reference



Momentum conservation laws at each annulus

➤ Axial force (thrust)

$$\frac{f_{axial(r)}}{\frac{1}{2}\rho U_{\infty}^2 2\pi r dr} = C_{T(r)} = 4a_{(r)}(1 - a_{(r)})$$

➤ Angular momentum (torque)

$$\frac{r \cdot f_{tangential(r)}}{\frac{1}{2}\rho U_{\infty}^2 r 2\pi r dr} = C_{Q(r)} = 4a'_{(r)}(1 - a_{(r)})\lambda \frac{r}{R}$$

$$\lambda = \frac{\Omega R}{U_{\infty}}$$

➤ Energy (power)

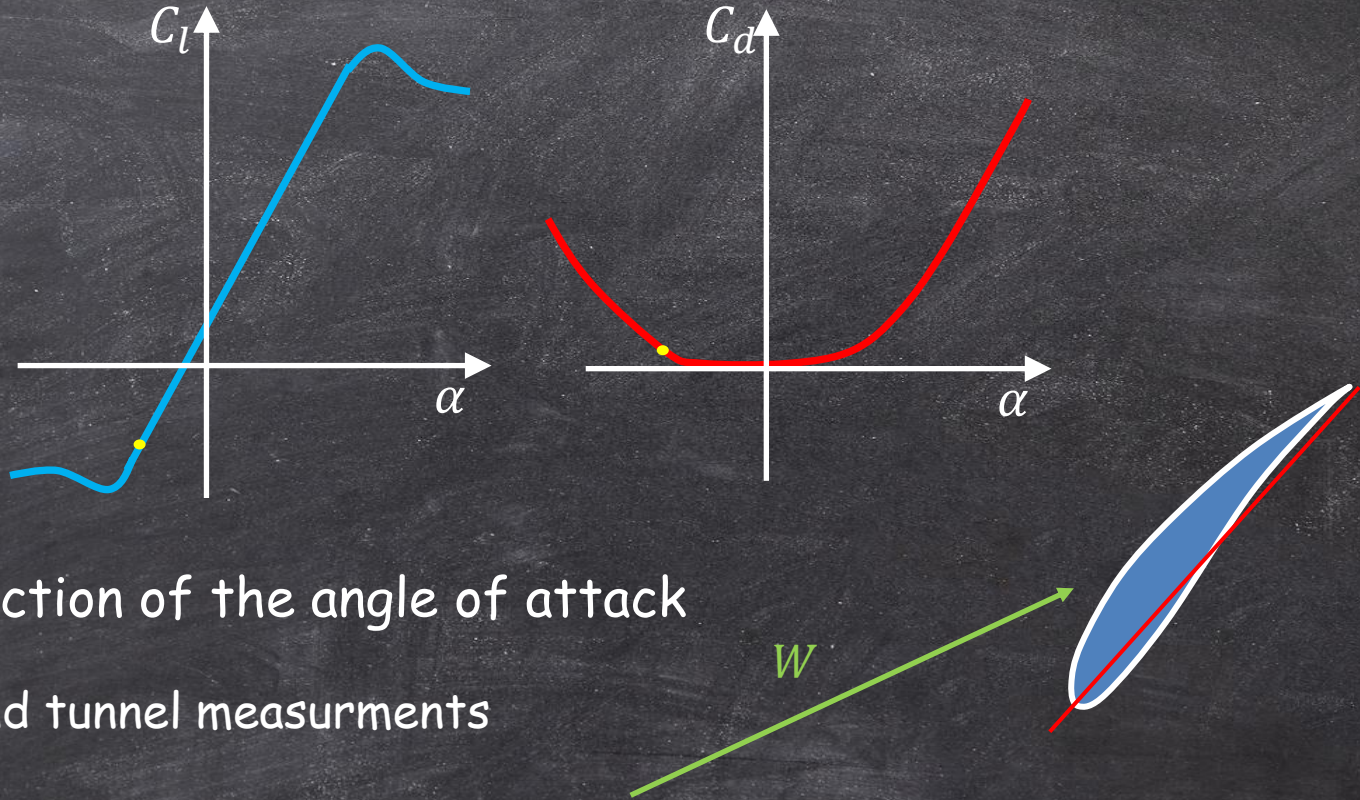
$$\frac{p_{(r)}}{\frac{1}{2}\rho U_{\infty}^3 2\pi r dr} = C_{P(r)} = 4a_{(r)}(1 - a_{(r)})^2$$

Blade element theory

- We use strip theory over the blade



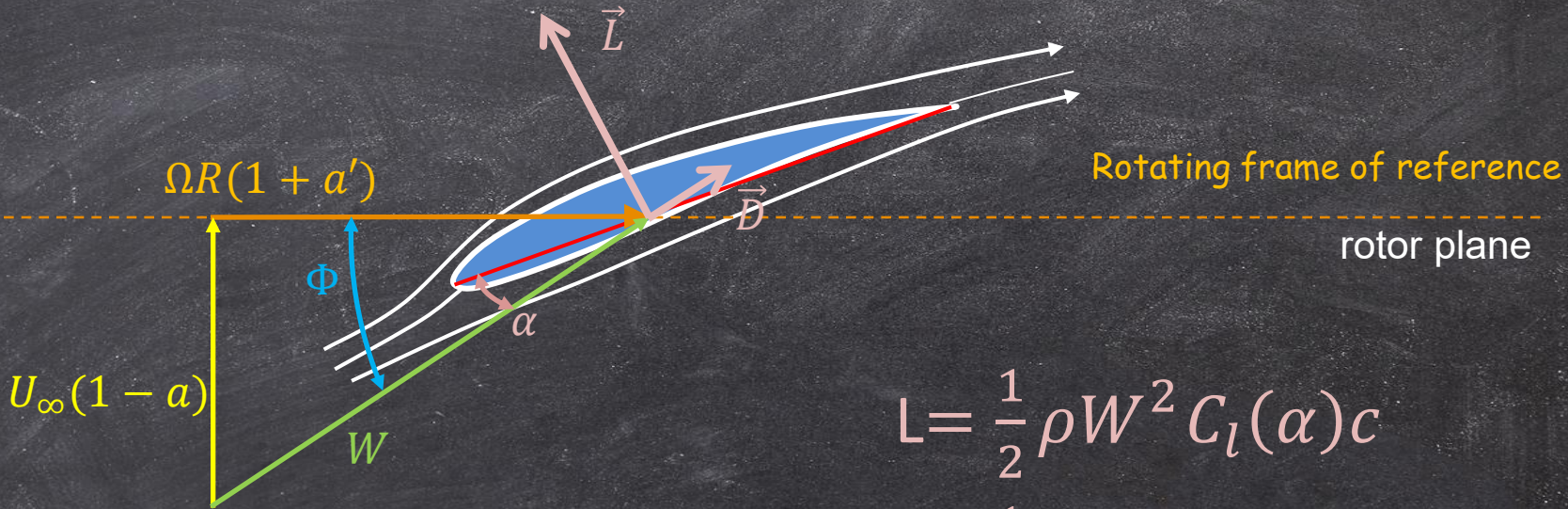
Airfoil data



C_l and C_d are function of the angle of attack

Normally from wind tunnel measurements

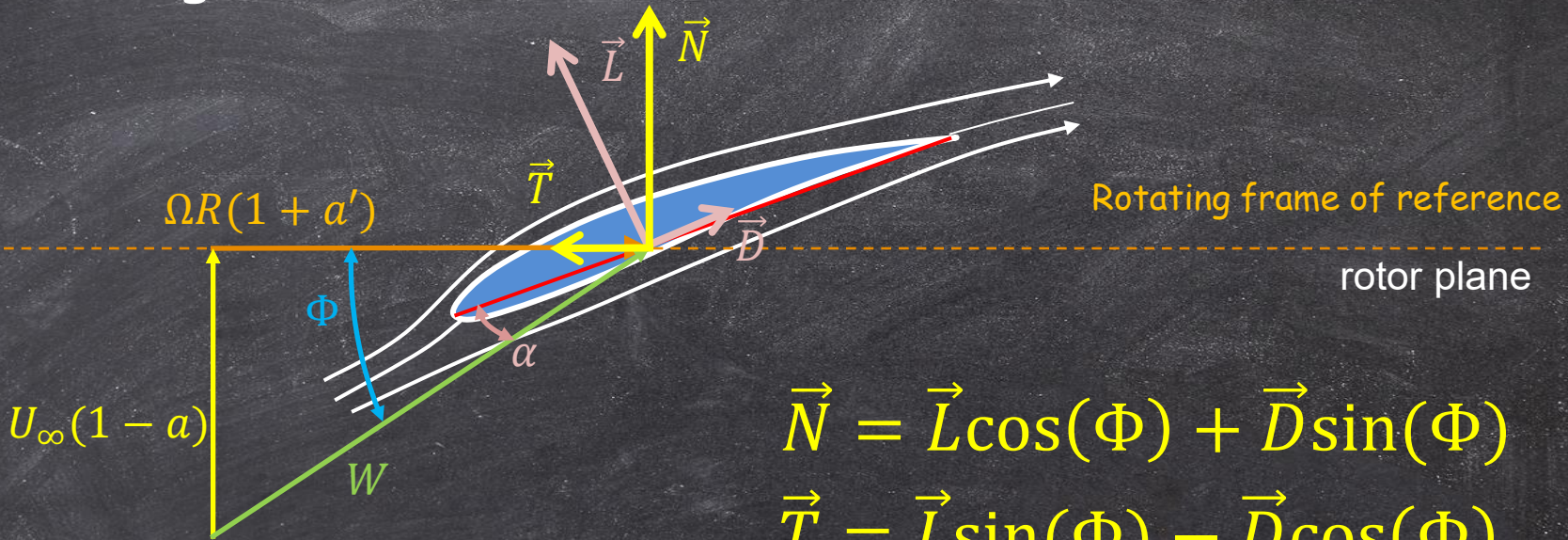
In strip theory, we consider the cross-section in a 2D flow



$$L = \frac{1}{2} \rho W^2 C_l(\alpha) c$$

$$D = \frac{1}{2} \rho W^2 C_d(\alpha) c$$

From lift and drag, we can calculate normal and tangential force for the momentum balance



$$\vec{N} = \vec{L}\cos(\Phi) + \vec{D}\sin(\Phi)$$

$$\vec{T} = \vec{L}\sin(\Phi) - \vec{D}\cos(\Phi)$$

(read: section 3.5 in *Wind Energy Handbook*,
Burton et al. 2011)

Momentum balance at each annulus

Axial force is responsible for rate of change of the axial momentum

$$\frac{1}{2} \rho W^2 B c (C_l \cos(\Phi) + C_d \sin(\Phi)) dr = \rho U_\infty (1 - a_{(r)}) 2\pi r dr 2 a_{(r)} U_\infty$$

Torque is responsible for rate of change of the angular momentum

$$\frac{1}{2} \rho W^2 B c (C_l \sin(\Phi) - C_d \cos(\Phi)) dr * r = \rho U_\infty (1 - a_{(r)}) 2\pi r dr * (\Omega r 2 a_{(r)})' * r$$

B = number of blades
c = chord length

BEM equations: iterative process

$$\frac{W^2}{U_\infty^2} B \frac{c}{R} (C_l \cos(\Phi) + C_d \sin(\Phi)) = 8\pi a(1-a)\mu \quad C_x$$

$$\frac{W^2}{U_\infty^2} B \frac{c}{R} (C_l \sin(\Phi) - C_d \cos(\Phi)) = 8\pi \lambda a'(1-a)\mu^2 \quad C_y$$

From
next
iteration

$$\begin{aligned} \frac{a}{1-a} &= \frac{\sigma_r}{4\sin^2\Phi} C_x \\ \frac{a'}{1+a'} &= \frac{\sigma_r}{4\sin\Phi\cos\Phi} C_y \end{aligned}$$

From
previous
iteration

$$\lambda = \frac{\Omega R}{U_\infty} \quad \text{tip speed ratio}$$

$$\sigma_r = \frac{Bc}{2\pi r} \quad \text{chord solidity}$$

$$\mu = \frac{r}{R}$$

Prandtl's correction for finite number of blades

- Because we have a finite number of blades with a finite bound circulation, a finite tip vortex is released at the blade tip
- This tip vortex creates an asymptotic induction field, for which our balance of momentum approach is no longer valid
- We need to correct the induction field for this tip (and also root effect)



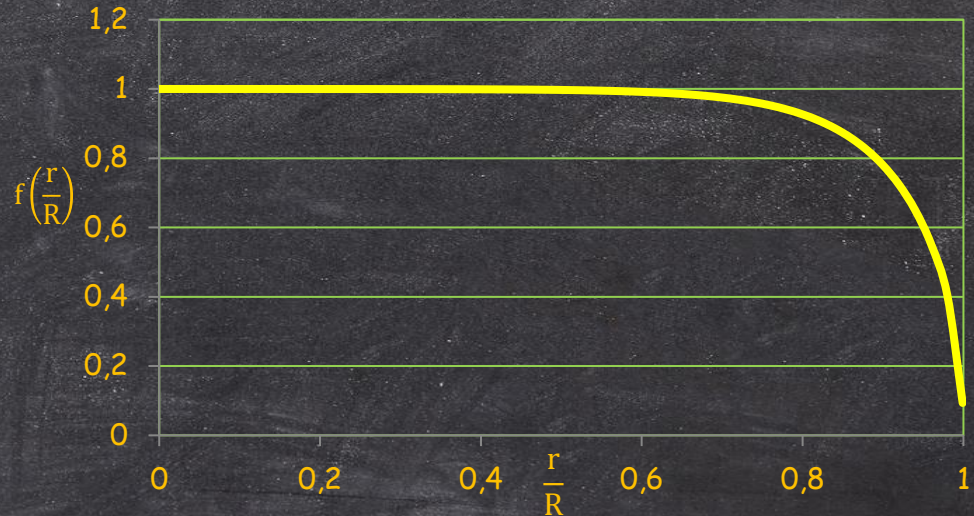
Prandtl's tip and root correction

Prandtl's correction applies to the induction term, both axial a and azimuthal induction a' . For application in the root region, one must mirror the equation.

$$f\left(\frac{r}{R}\right) = \frac{2}{\pi} \arccos(e^{-\pi(1-r/R)/d})$$

$$d = \frac{2\pi}{B} \frac{(1-a)}{\sqrt{\lambda^2 + (1-a)^2}}$$

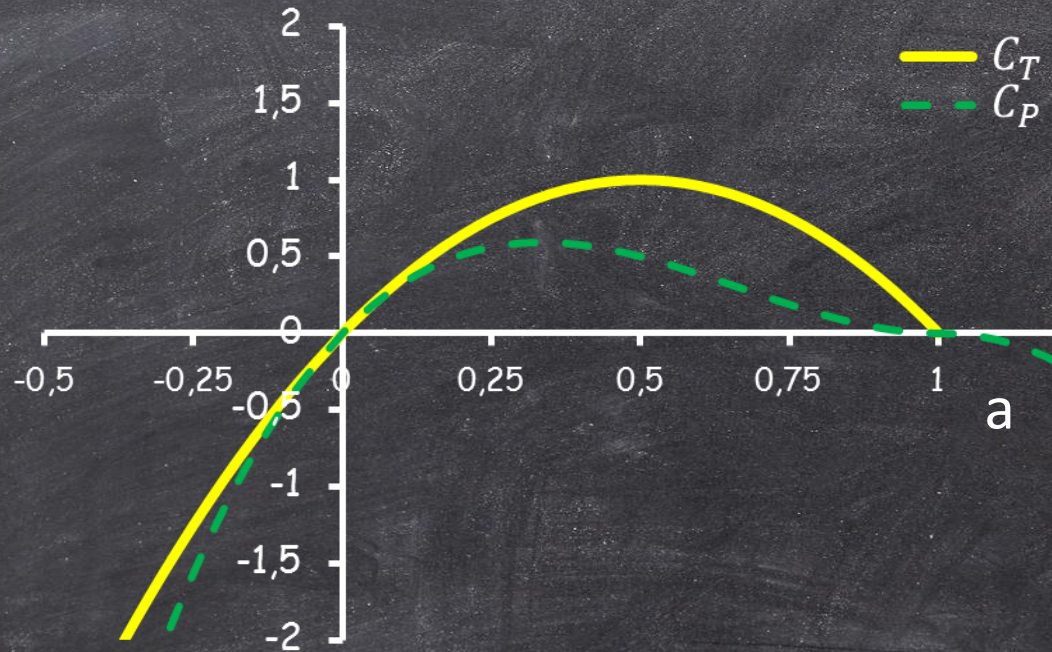
$$\Phi = \operatorname{atan}\left(\frac{1}{\lambda r/R} \frac{(1 - a/f)}{(1 + a'/f)}\right)$$



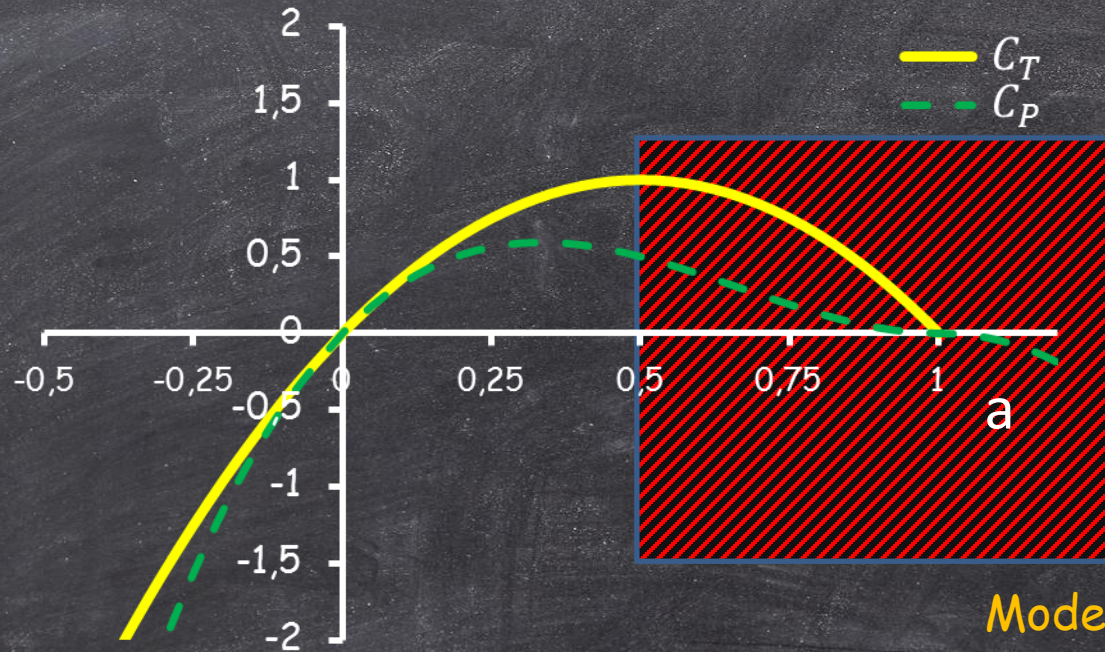
Correction for heavily loaded streamtubes

- For values of induction $a > 0.5$, the streamtube models results in flow reversal and breakdown for the continuity equation.
- We must therefore correct the model of thrust coefficient as a function of induction

C_T and C_P as a function of a



C_T and C_P as a function of a



Model is invalid

Glauert's correction for heavily loaded rotors

$$C_t = \begin{cases} 4a(1-a), & a < 1 - \frac{\sqrt{C_{T1}}}{2} \\ C_{T1} - 4(\sqrt{C_{T1}} - 1)(1-a), & a \geq 1 - \frac{\sqrt{C_{T1}}}{2} \end{cases}$$

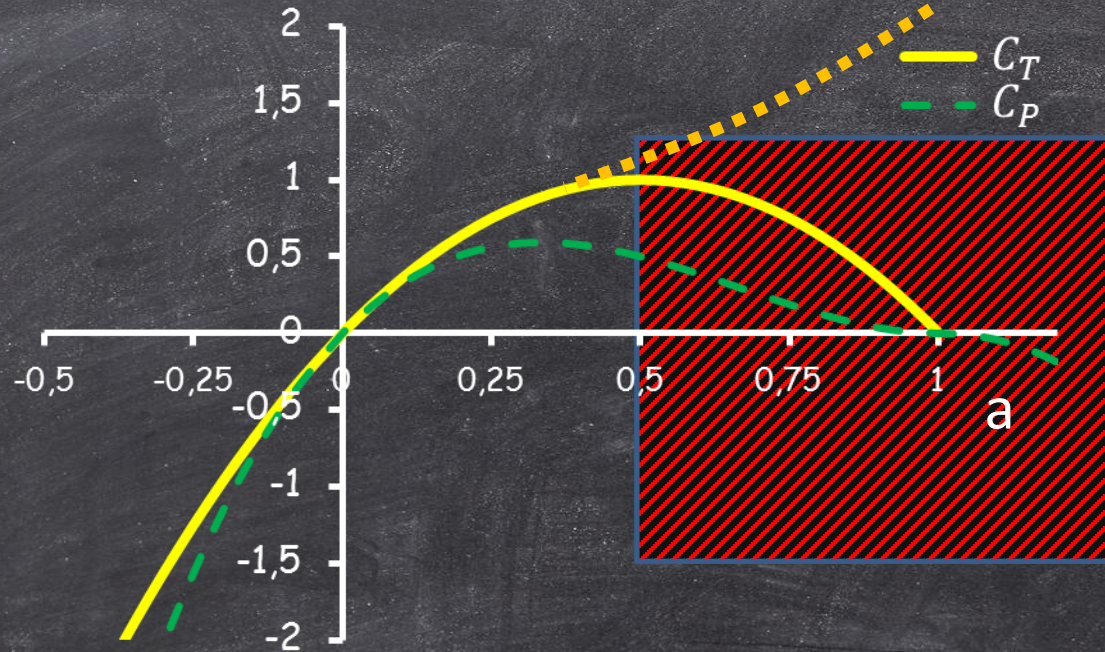
$$a = \begin{cases} \frac{1}{2} - \frac{\sqrt{1-C_T}}{2}, & C_T < C_{T2} \\ 1 + \frac{C_T - C_{T1}}{4\sqrt{C_{T1}} - 4}, & C_T \geq C_{T2} \end{cases}$$

$$C_{T1} = 1.816 \text{ and } C_{T2} = 2\sqrt{C_{T1}} - C_{T1}$$

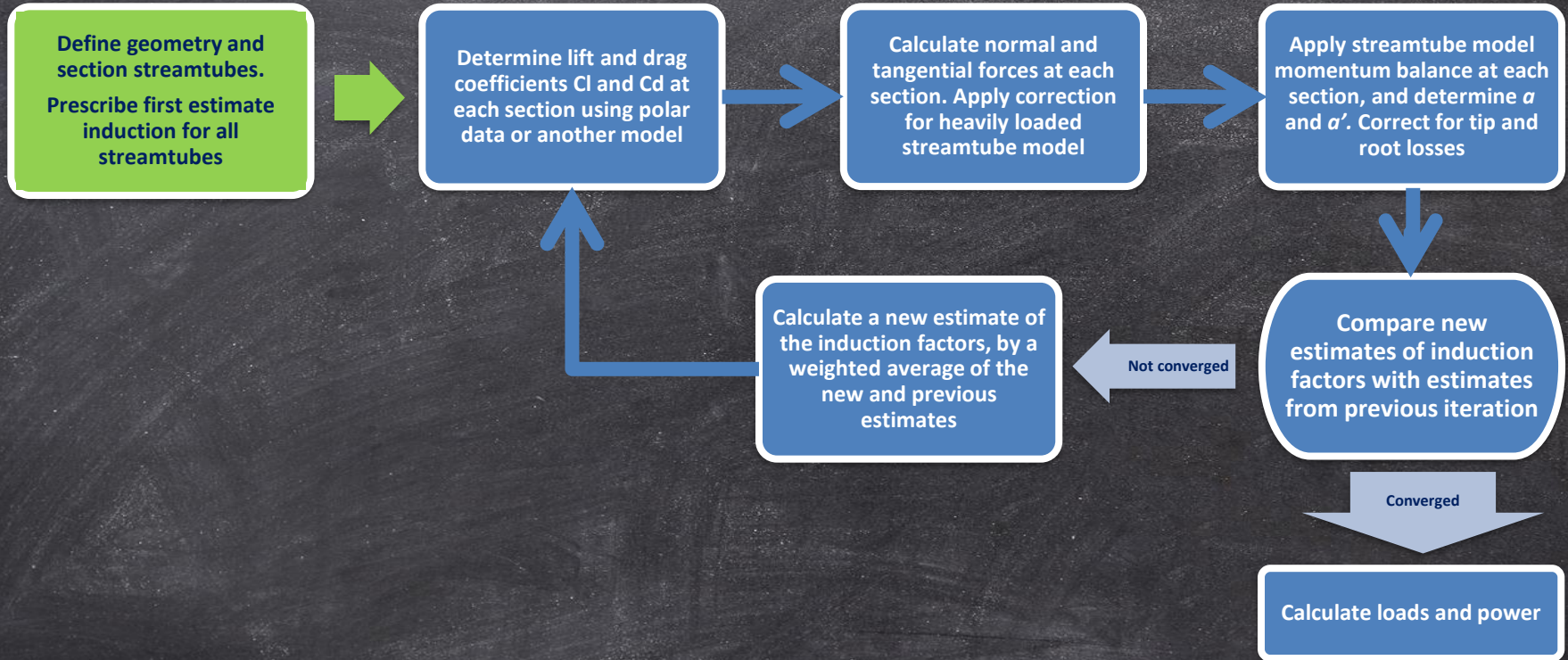
C_T is corrected for $a > a_1$

Glauert correction

$$C_T = C_{T1} - 4(\sqrt{C_{T1}} - 1)(1 - a), \text{ with } C_{T1} = 1,816$$



BEM algorithm



Today we:

- Discussed the key points to setup a BEM model
- Discussed the effect of torque and wake rotation
- Discussed main corrections for heavily loaded streamtubes and tip/root losses
- Defined an algorithm for a Blade Element Momentum model

Rotor and Wake Aerodynamics

Models for rotors in steady yaw

In the previous modules, we have:

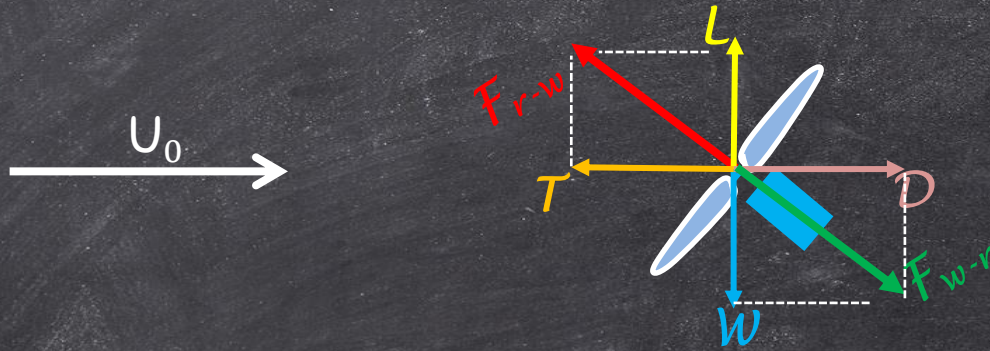
- Derived BEM of an axial rotor;
- Defined an algorithm for setting up a BEM model;
- Discussed the main corrections for heavily loaded streamtubes and tip/root losses;

Our objective for today

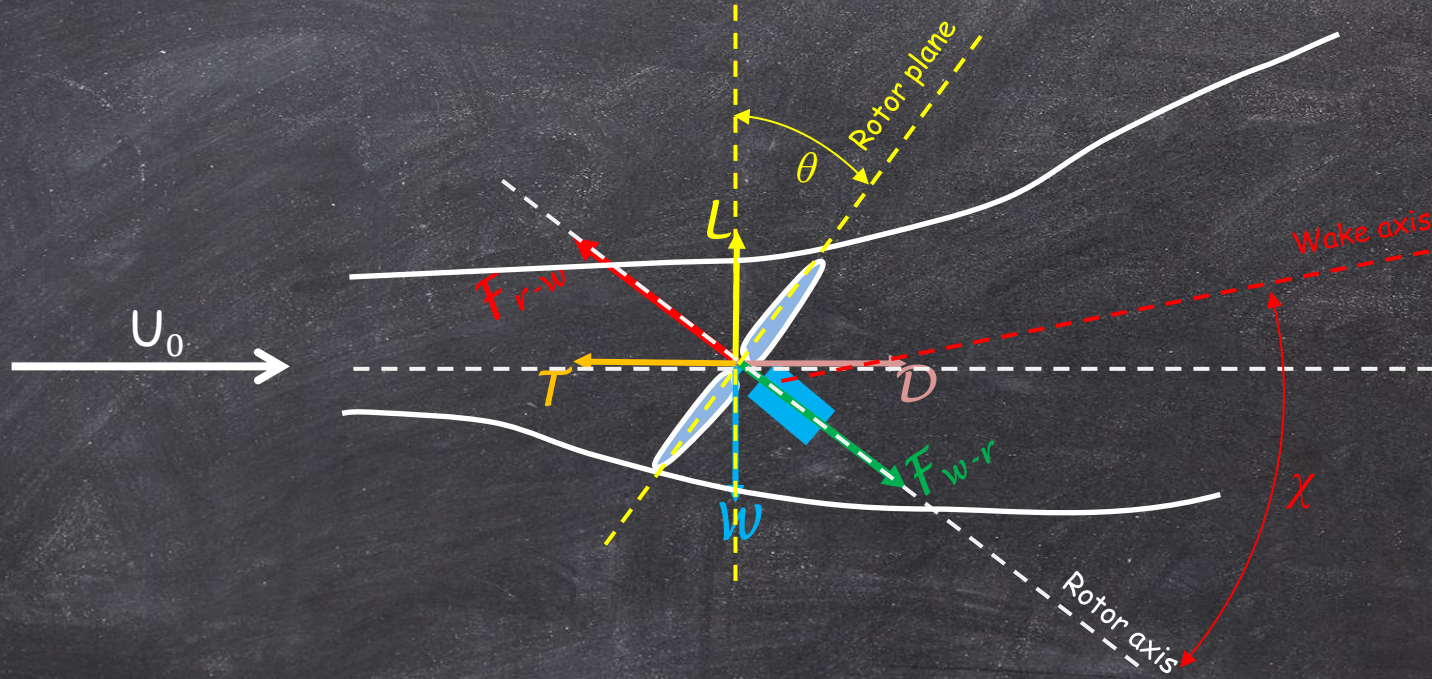
By the end of this module, you should be able to

- Apply **Momentum Theory, Glauert Theory and vortex cylinder model** to a horizontal axis rotor in yaw or a helicopter in forward flight.

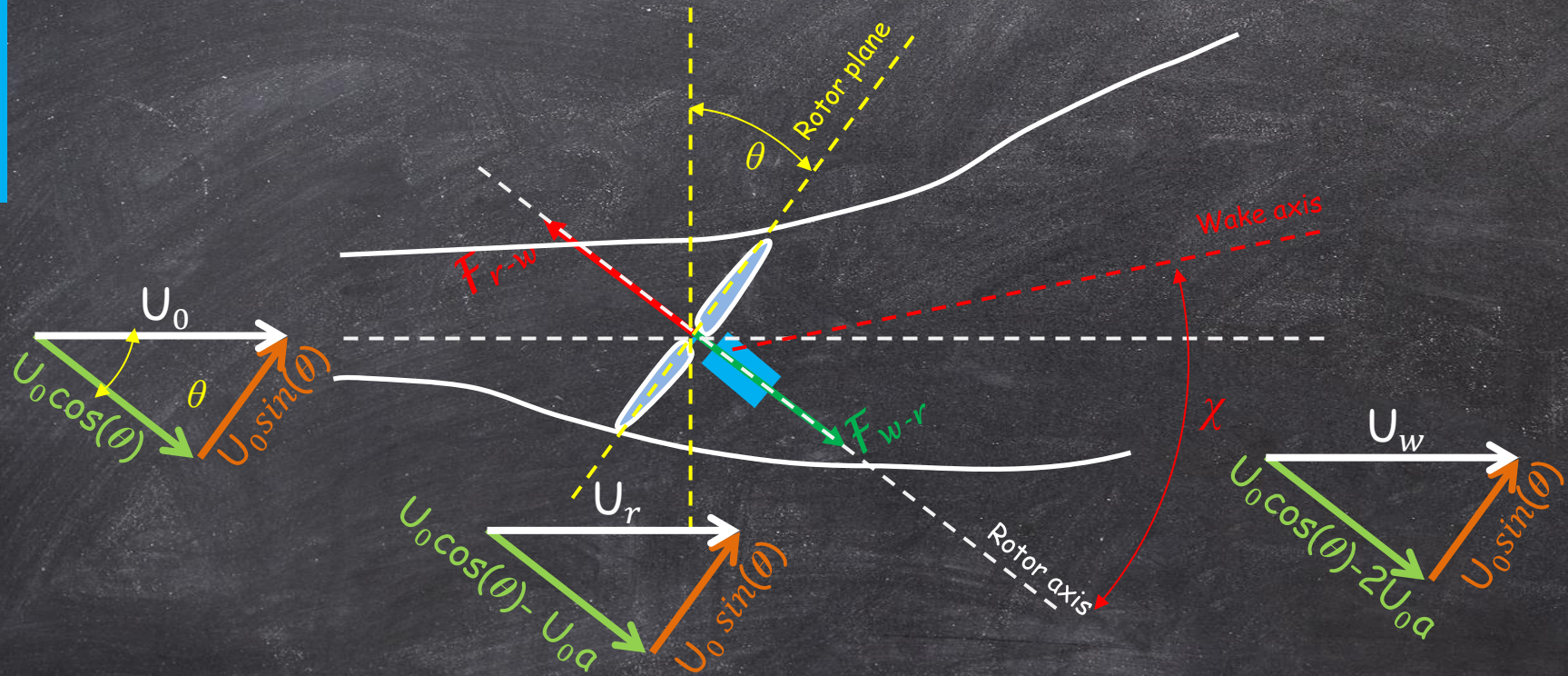
Momentum theory for yawed condition



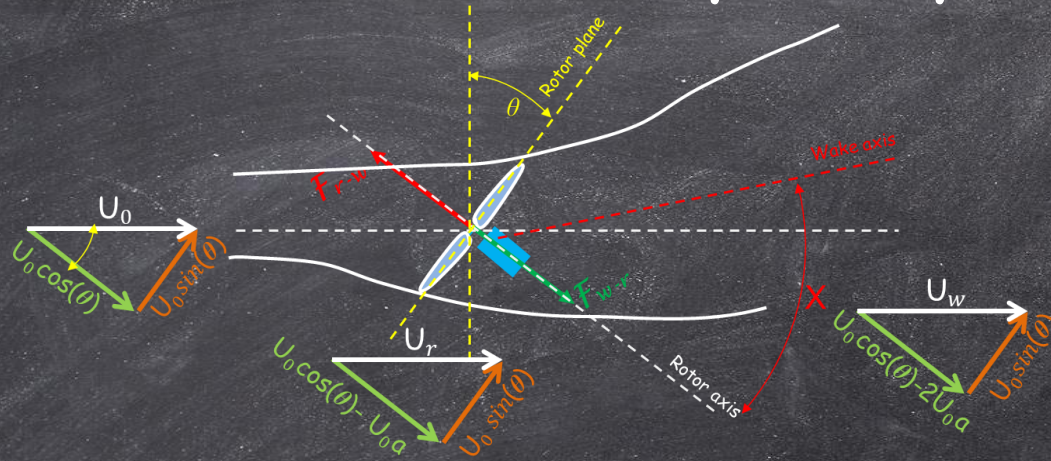
Momentum theory for yawed condition



Momentum theory for yawed condition



Momentum theory for yawed condition



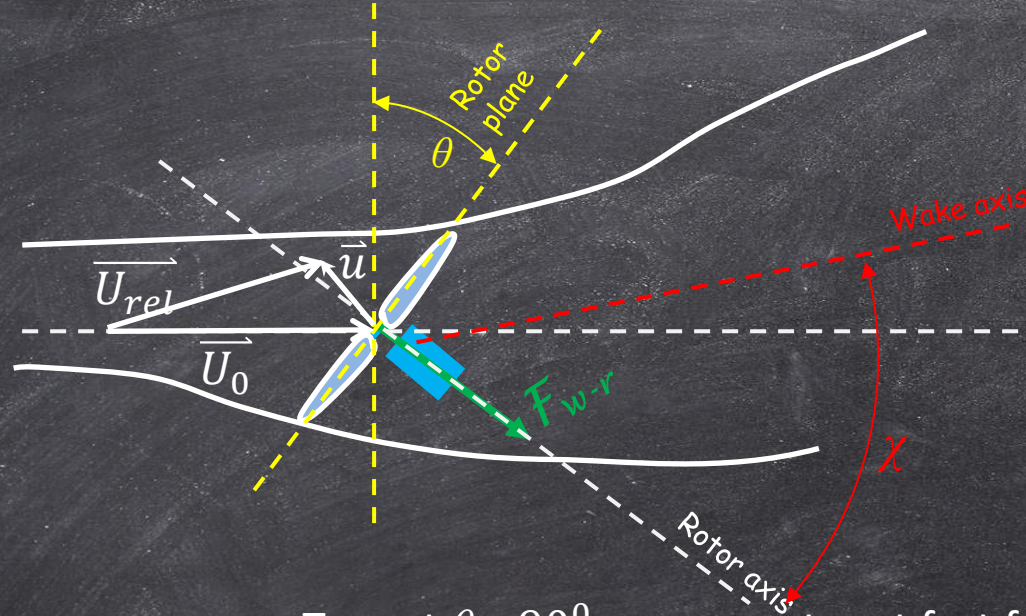
$$C_T = \frac{F}{\frac{1}{2}\rho A_d U_0^2} = 4a(\cos\theta - a)$$

$$C_P = \frac{P}{\frac{1}{2}\rho A_d U_0^3} = 4a(\cos\theta - a)^2$$

$$F = \rho A_d U_0 (\cos\theta - a) * (U_0 \cos\theta - (U_0 \cos\theta - 2aU_0)) = \rho A_d U_0^2 (\cos\theta - a) 2a$$

$$P = F U_d = F U_0 (\cos\theta - a) = \rho A_d U_0^3 (\cos\theta - a)^2 2a$$

Glauert theory for yawed condition



The lifting line theory gives

$$u = \frac{2F}{\pi(2R)^2 \rho U_{rel}}$$

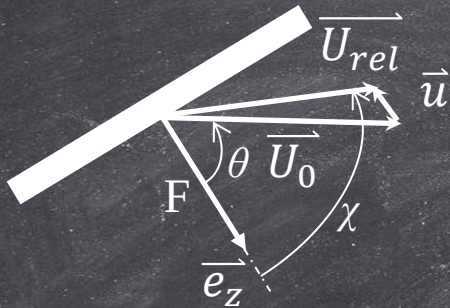
$$\vec{U}_{rel} = \vec{U}_0 + \vec{u}$$

True at $\theta = 90^\circ$, a gyrocopter at fast forward flight

True at $\theta = 0^\circ$, axial MT of aligned wind turbine

assumed to be true for all θ

Glauert theory for yawed condition



At high θ , F and u are assumed to be normal to the rotor plane

$$U_{rel} = \sqrt{(U_0 \cos \theta - u)^2 + (U_0 \sin \theta)^2}$$

$$F = \rho \pi R^2 2u \sqrt{(U_0 \cos \theta - u)^2 + (U_0 \sin \theta)^2}$$

$$P = \rho \pi R^2 2u \sqrt{(U_0 \cos \theta - u)^2 + (U_0 \sin \theta)^2} (U_0 \cos \theta - u)$$

$$C_T = 4a \sqrt{1 - a(2 \cos \theta - a)}$$

$$C_P = 4a \sqrt{1 - a(2 \cos \theta - a)} (\cos \theta - a)$$

Glauert theory for yawed condition

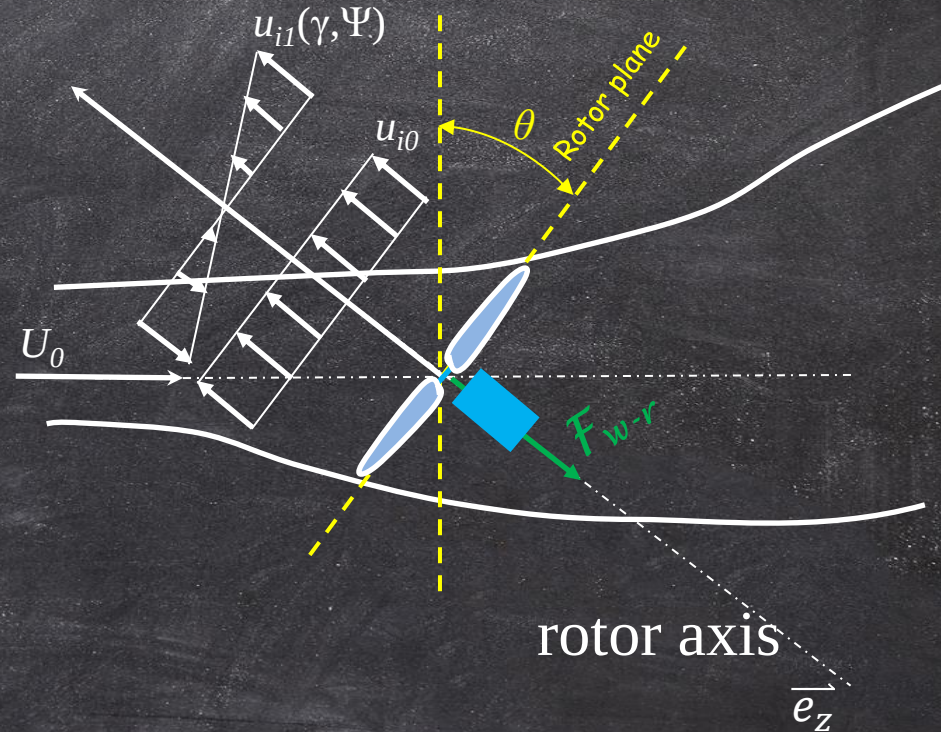
$$u = u_{i0} \left(1 + K(\chi) \frac{r}{R} \sin \Psi \right)$$

$$u_{i0} = aU_0$$

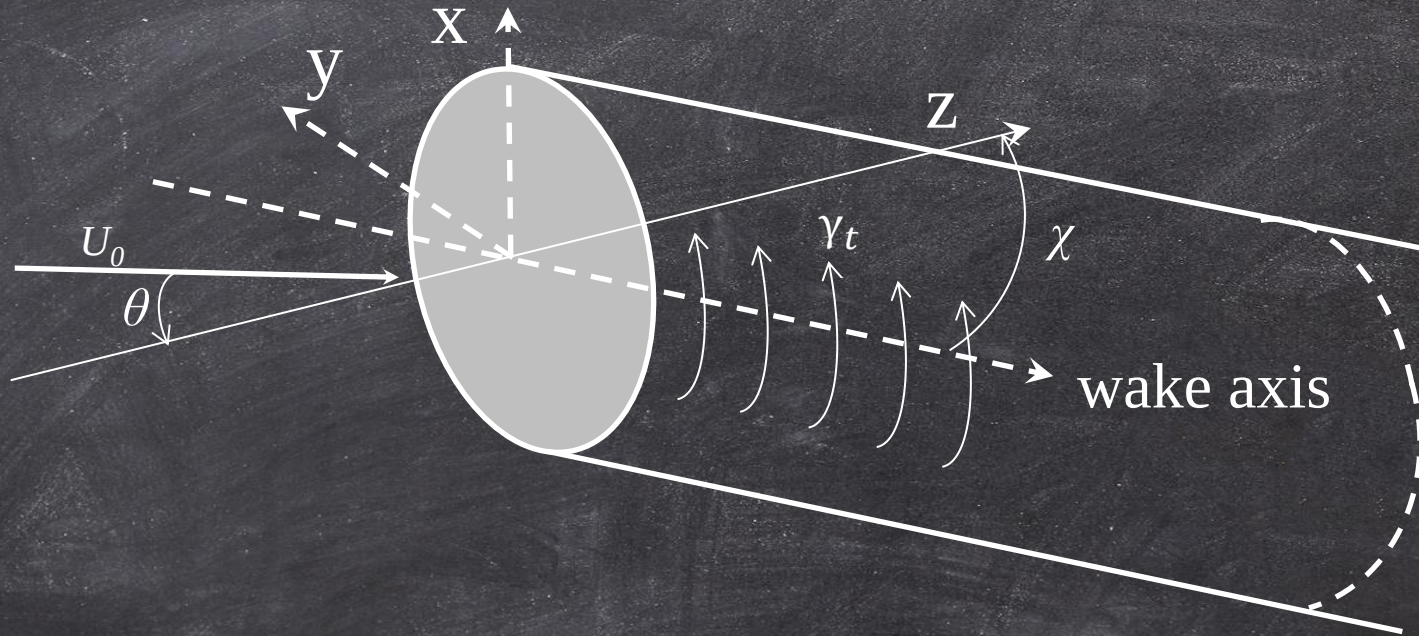
$$u_{i1}(\theta, \Psi) = u_{i0} K(\chi) \frac{r}{R} \sin \Psi$$

ψ --- the azimuthal position

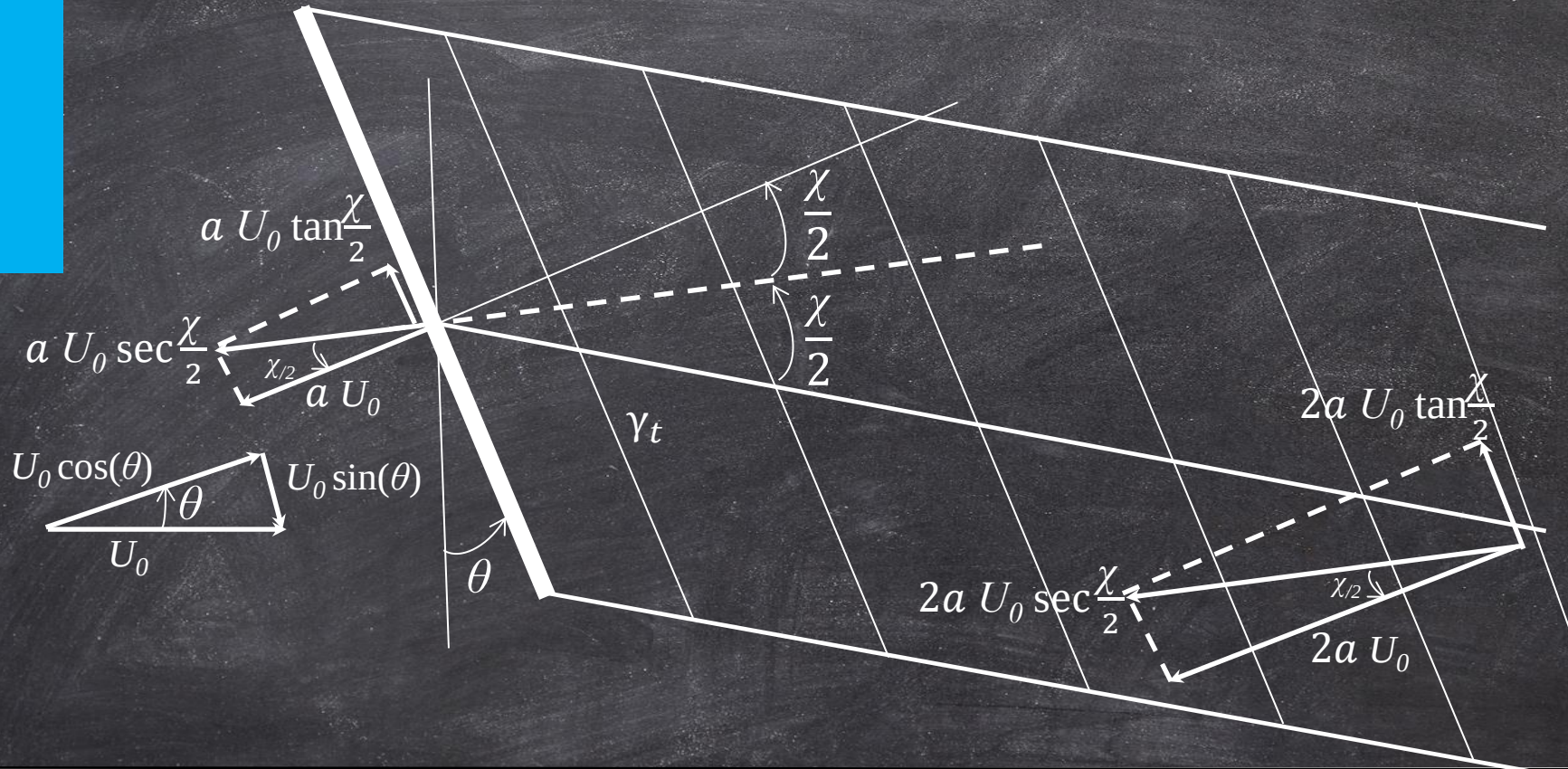
K --- depends on the yaw angles



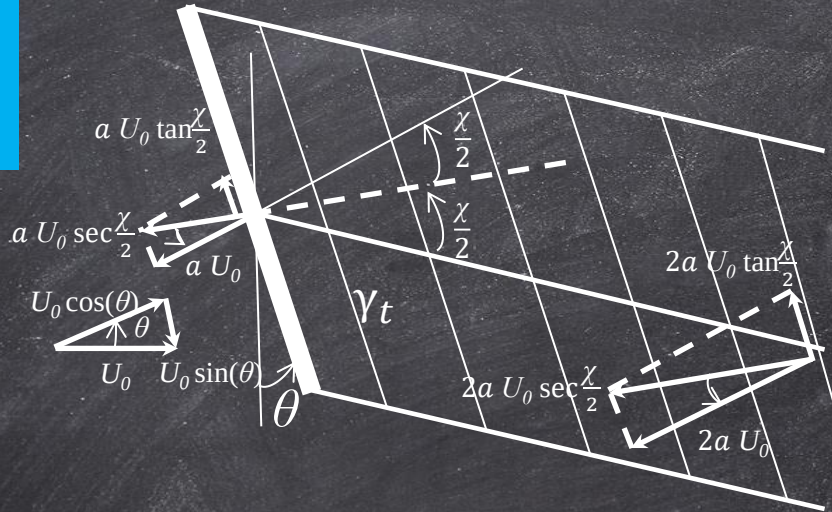
Vortex cylinder model for yawed condition



Vortex cylinder model for yawed condition



Vortex cylinder model for yawed condition



The velocity component at the disc defines the skew angle

$$\tan \chi = \frac{U_0 (\sin \theta - a \tan \frac{\chi}{2})}{U_0 (\cos \theta - a)}$$

$$\tan \chi = \frac{2 \tan \frac{\chi}{2}}{1 - \tan^2 \frac{\chi}{2}}$$

A close approximation is given by
 $\chi = (0.6a + 1)\theta$

Vortex cylinder model for yawed condition

'upstream' Bernoulli equation :

$$p_0 + \frac{1}{2}\rho U_0^2 = p_d^+ + \frac{1}{2}\rho U_d^2$$

'downstream' Bernoulli equation :

$$p_d^- + \frac{1}{2}\rho U_d^2 = p_0 + \frac{1}{2}\rho U_0^2 [(\cos\theta - 2a)^2 + (\sin\theta - 2a \tan \frac{\chi}{2})^2]$$

$$C_T = \frac{\Delta p}{\frac{1}{2}\rho U_0^2} = 4a(\cos\theta + \sin\theta \tan \frac{\chi}{2} - a \sec^2 \frac{\chi}{2})$$

$$C_P = 4a(\cos\theta + \sin\theta \tan \frac{\chi}{2} - a \sec^2 \frac{\chi}{2})(\cos\theta - a)$$

Vortex cylinder model for yawed condition

Coleman derived the axial velocity at the disc using the vortex cylinder

$$u \approx u_{i0} \left(1 + 2 \tan \frac{\chi}{2} F\left(\frac{r}{R}\right) \sin \Psi \right)$$

It takes the same format as Glauert theory: $u = u_{i0} \left(1 + K(\chi) \frac{r}{R} \sin \Psi \right)$

Coleman's model gives $K(\chi) = 2 \tan \frac{\chi}{2}$

In this module, we have discussed

- Axial momentum theory for a rotor in yaw.
- Glauert theory for a rotor in yaw.
- Vortex cylinder model for a rotor in yaw.

Reference

1. T. Burton, N. Jenkins, D. Sharpe, and E. Bossanyi. Wind energy handbook. John Wiley & Sons, 2011.
2. J.G. Schepers. Engineering models in wind energy aerodynamics, PhD thesis, Delft University of Technology, 2012.
3. E. Branlard. Wind Turbine Aerodynamics and Vorticity-Based Methods, 2017.